

NPTEL Course on "Computer System Optimization for ML"

Lecture #60

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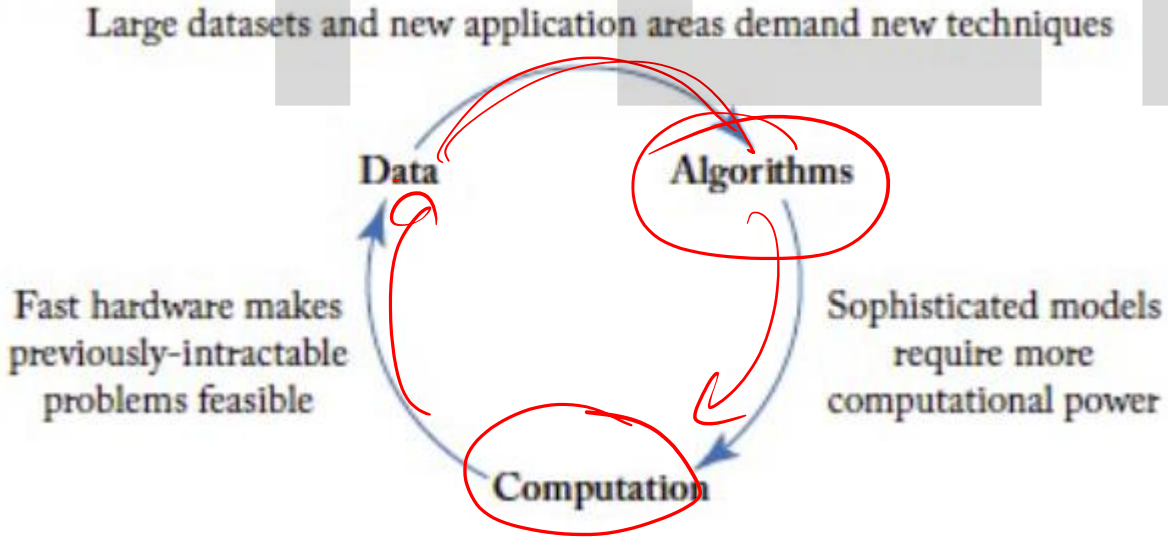
Agenda

- **Recap of the course**

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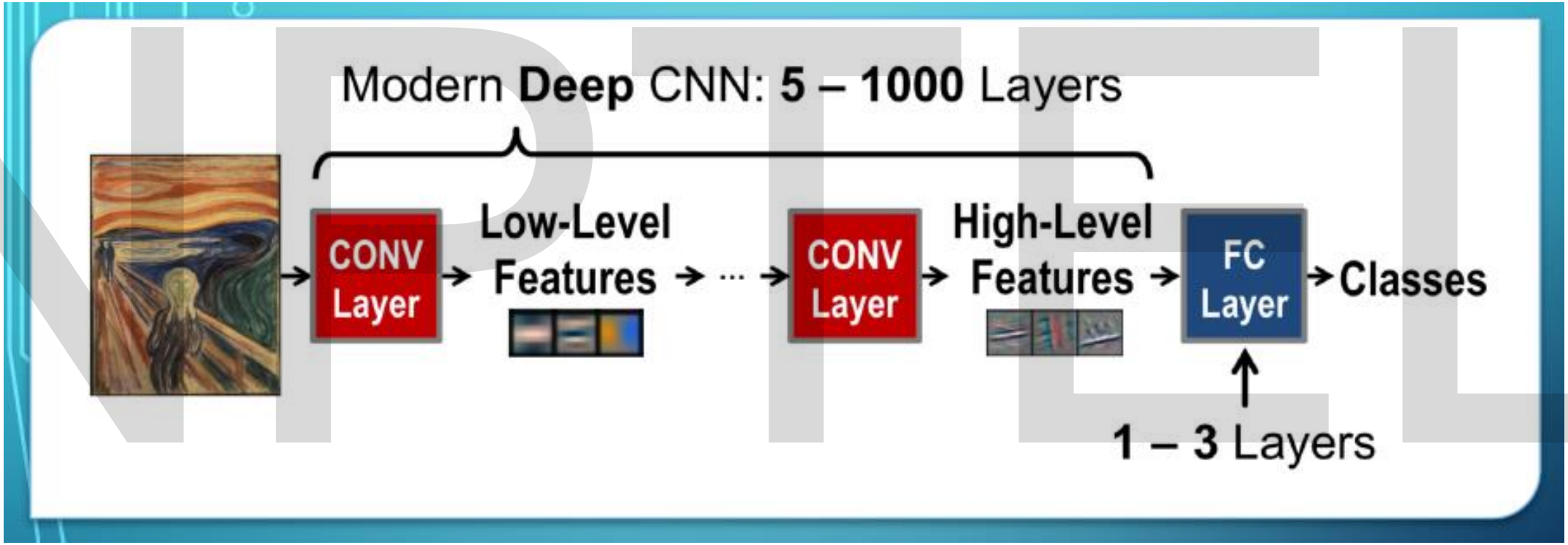
What Actually Changed?

- **Large datasets and new application area demand new techniques**
  - Autonomous driving
- **Sophisticated models require more computational power**
  - LR < RT/DT < DNN < CNN < GNN < LSTM < Transformer ….
- **Fast hardware makes previously-intractable problems feasible**
  - CPU < GPU < TPU < Accelerators …..
- **It's a cycle**

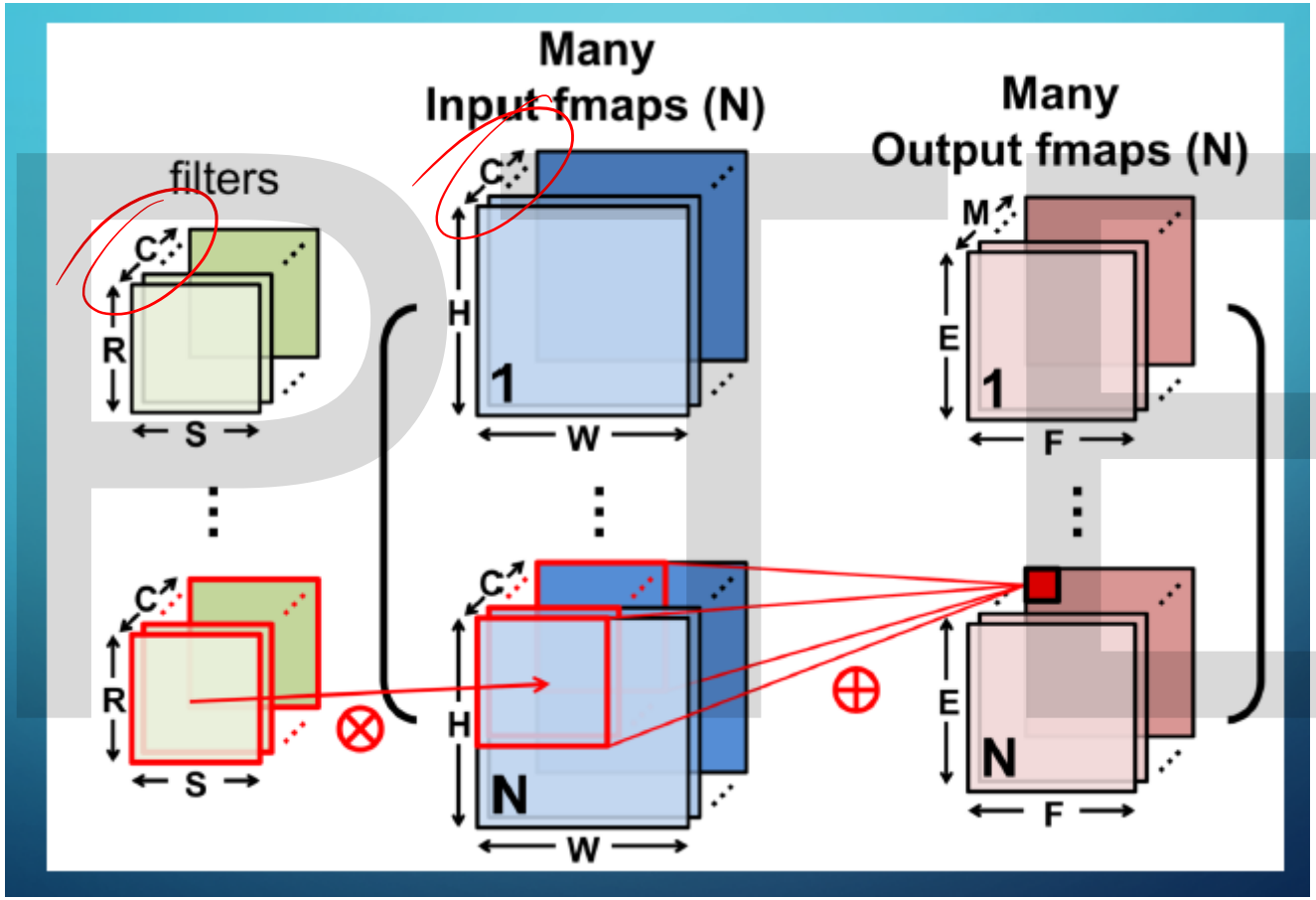


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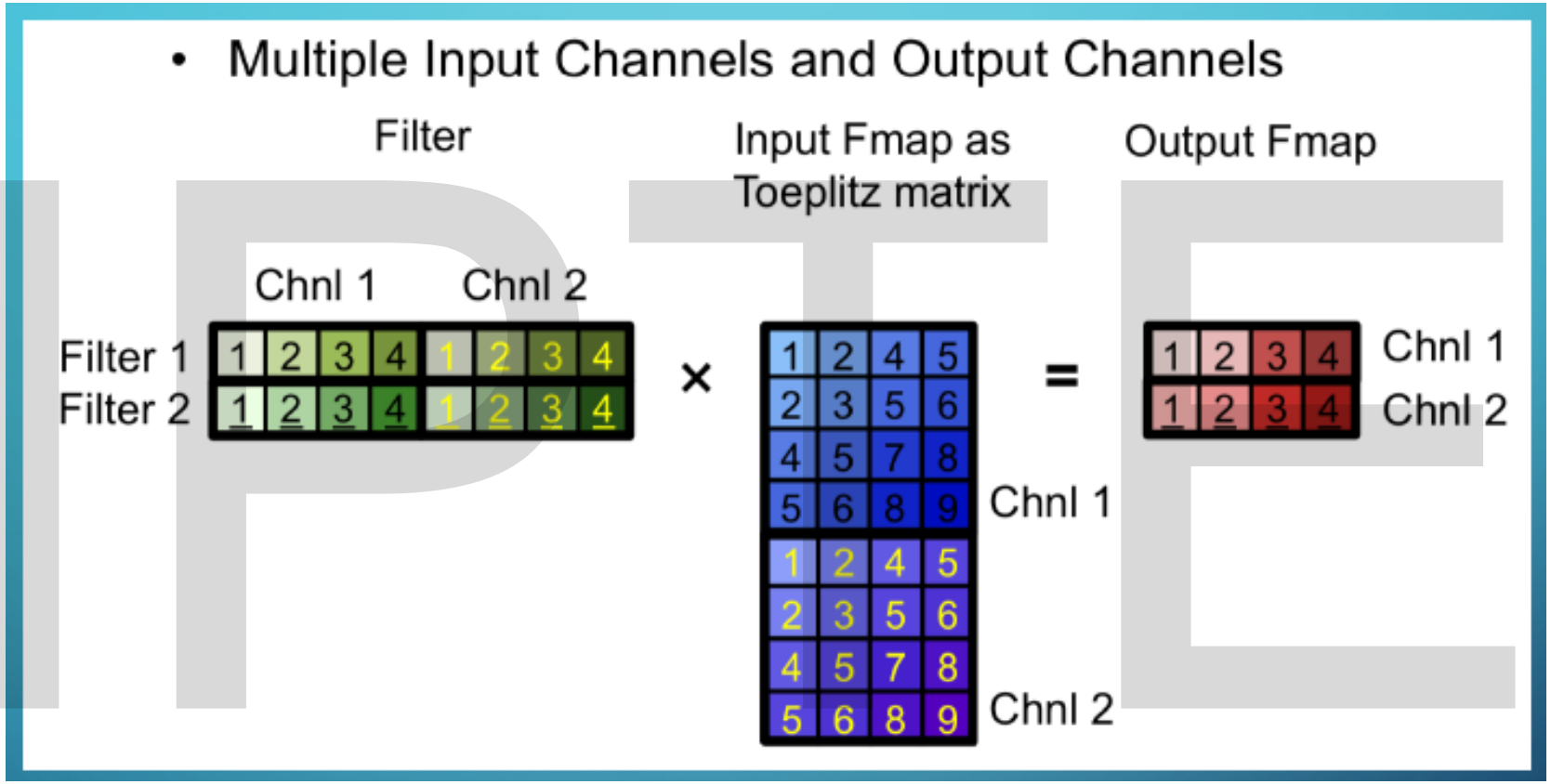
Deep CNNs


Modern **Deep** CNN: **5 – 1000** Layers
CONV Layer
Low-Level Features
CONV Layer
High-Level Features
FC Layer
Classes
1 – 3 Layers

Conv Layer (Batch Multiple Inputs)



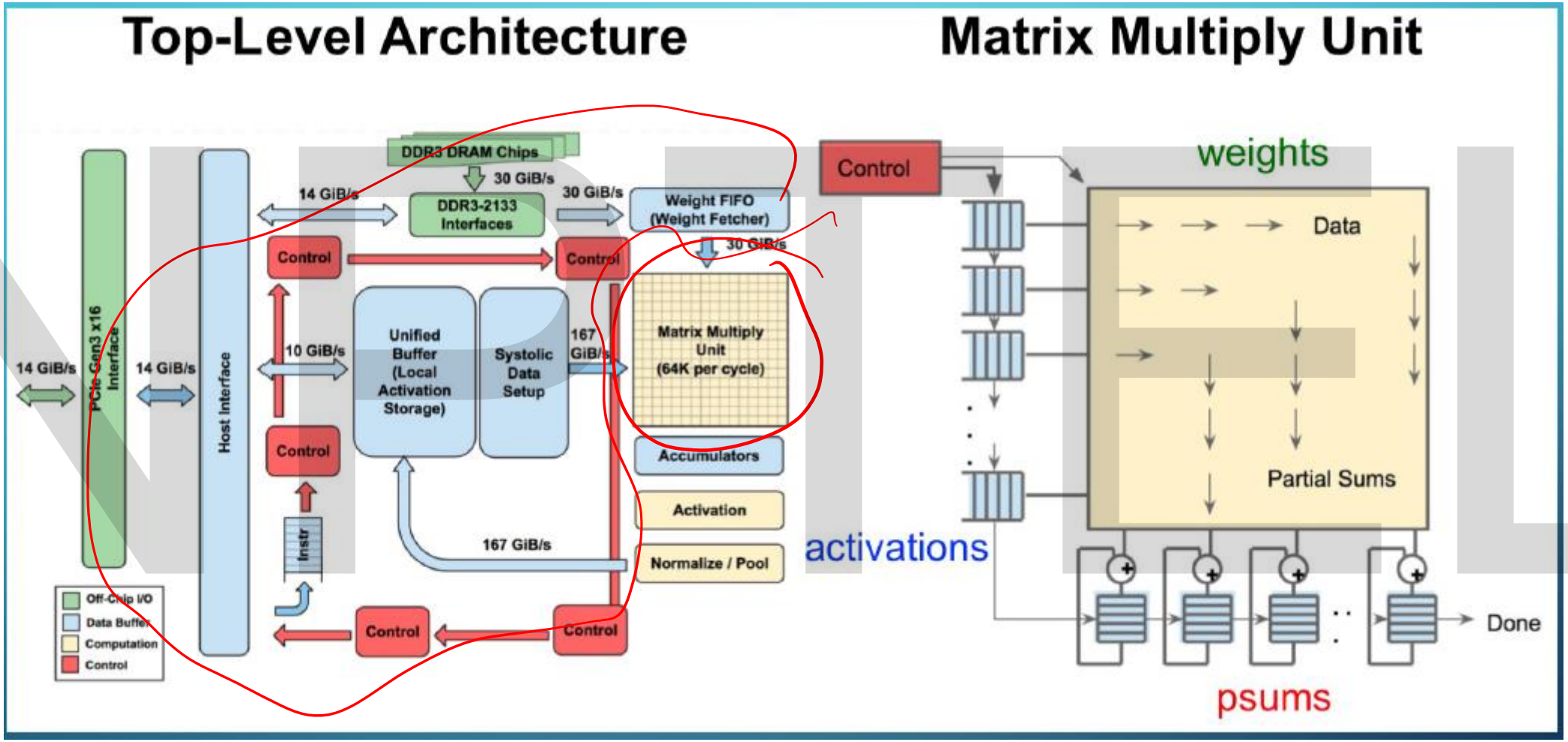
Conv Layer Computation



- **We can flatten both filters and input feature maps in the same way as before**
- **Naturally generate flattened output feature maps of two channels**

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Tensor Processing Unit

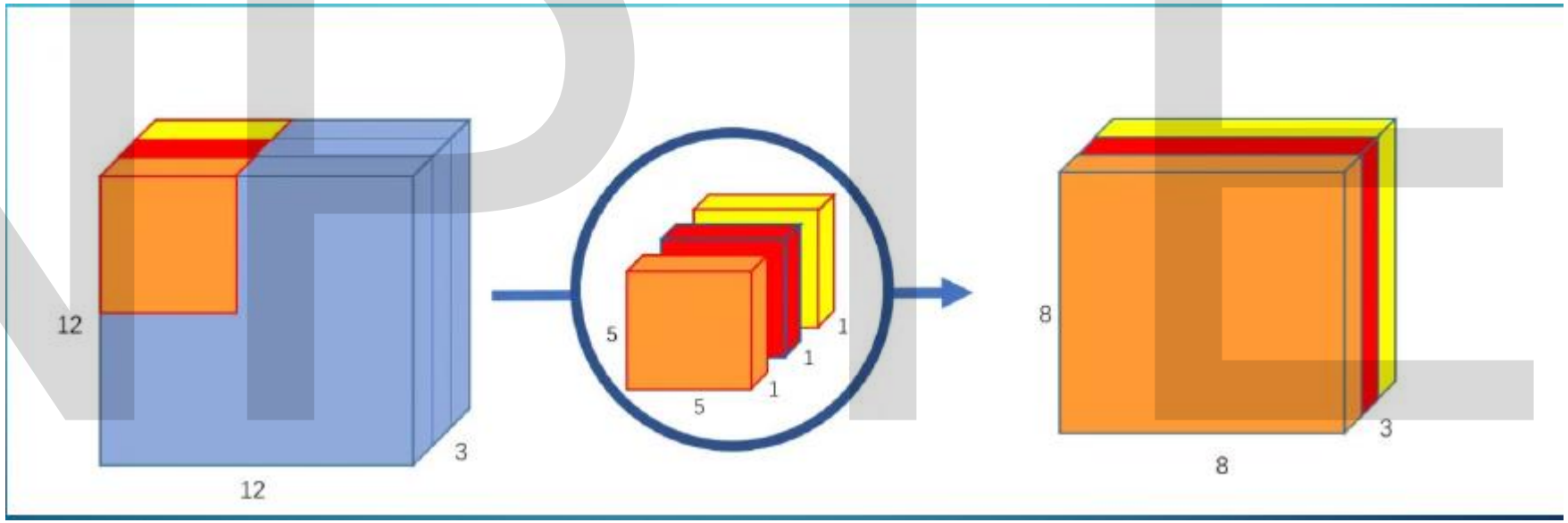


Summary of Popular DNNs

Metrics	LeNet-5	AlexNet	VGG-16	GoogLeNet (v1)	ResNet-50
Top-5 error	n/a	16.4	7.4	6.7	5.3
Input Size	28x28	227x227	224x224	224x224	224x224
# of CONV Layers	**2**	**5**	**16**	**21 (depth)**	**49**
Filter Sizes	5	3, 5,11	3	1, 3 , 5, 7	1, 3, 7
# of Channels	1, 6	3 - 256	3 - 512	3 - 1024	3 - 2048
# of Filters	6, 16	96 - 384	64 - 512	64 - 384	64 - 2048
Stride	1	1, 4	1	1, 2	1, 2
# of Weights	2.6k	2.3M	14.7M	6.0M	23.5M
# of MACs	283k	666M	15.3G	1.43G	3.86G
# of FC layers	**2**	**3**	**3**	**1**	**1**
# of Weights	58k	58.6M	124M	1M	2M
# of MACs	58k	58.6M	124M	1M	2M
Total Weights	**60k**	**61M**	**138M**	**7M**	**25.5M**
Total MACs	**341k**	**724M**	**15.5G**	**1.43G**	**3.9G**

Depth-wise Separable Convolution

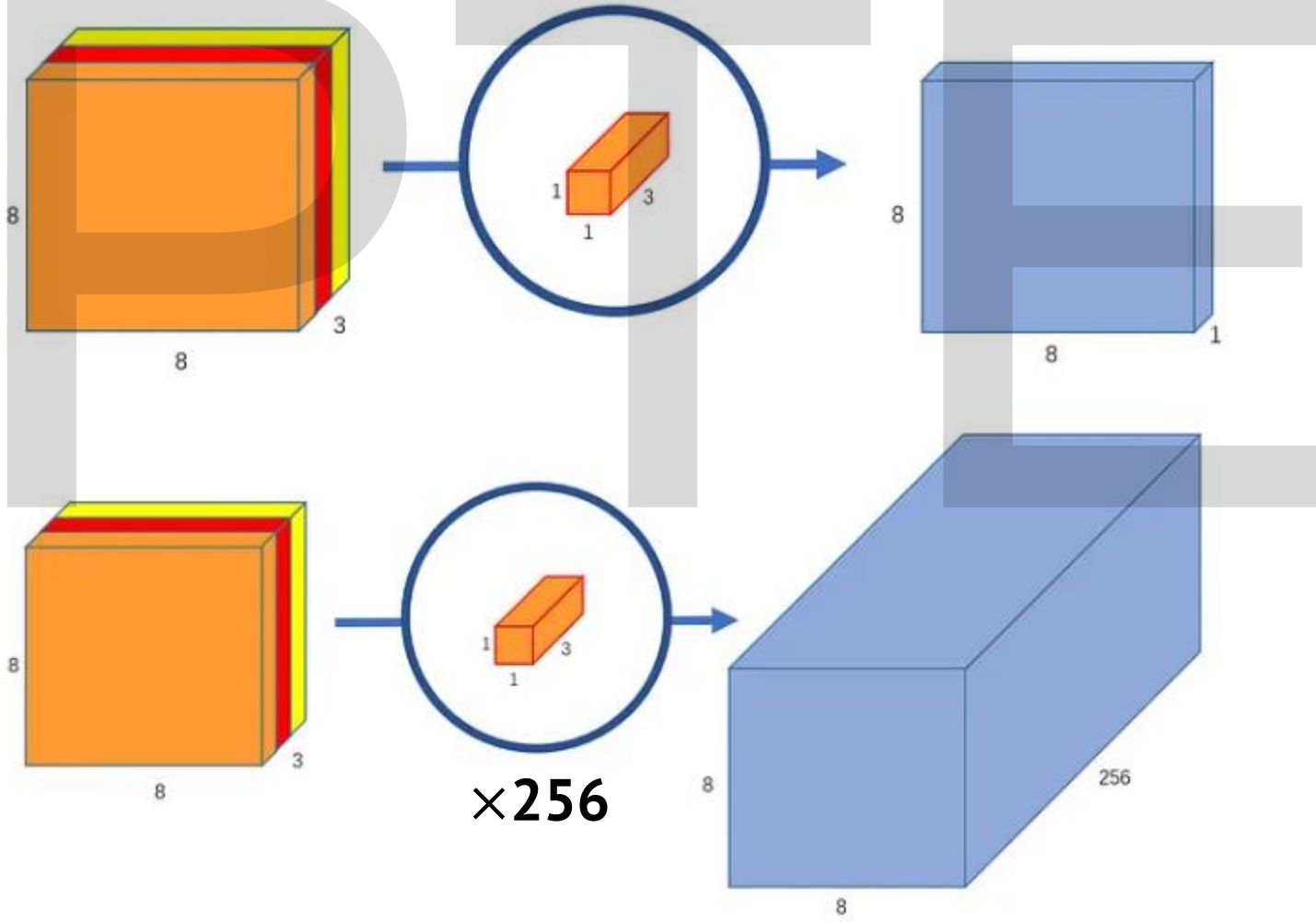
- **Step-1: Depth-wise Convolution**



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Depth-wise Separable Convolution

- **Step-2: Point-wise Convolution**



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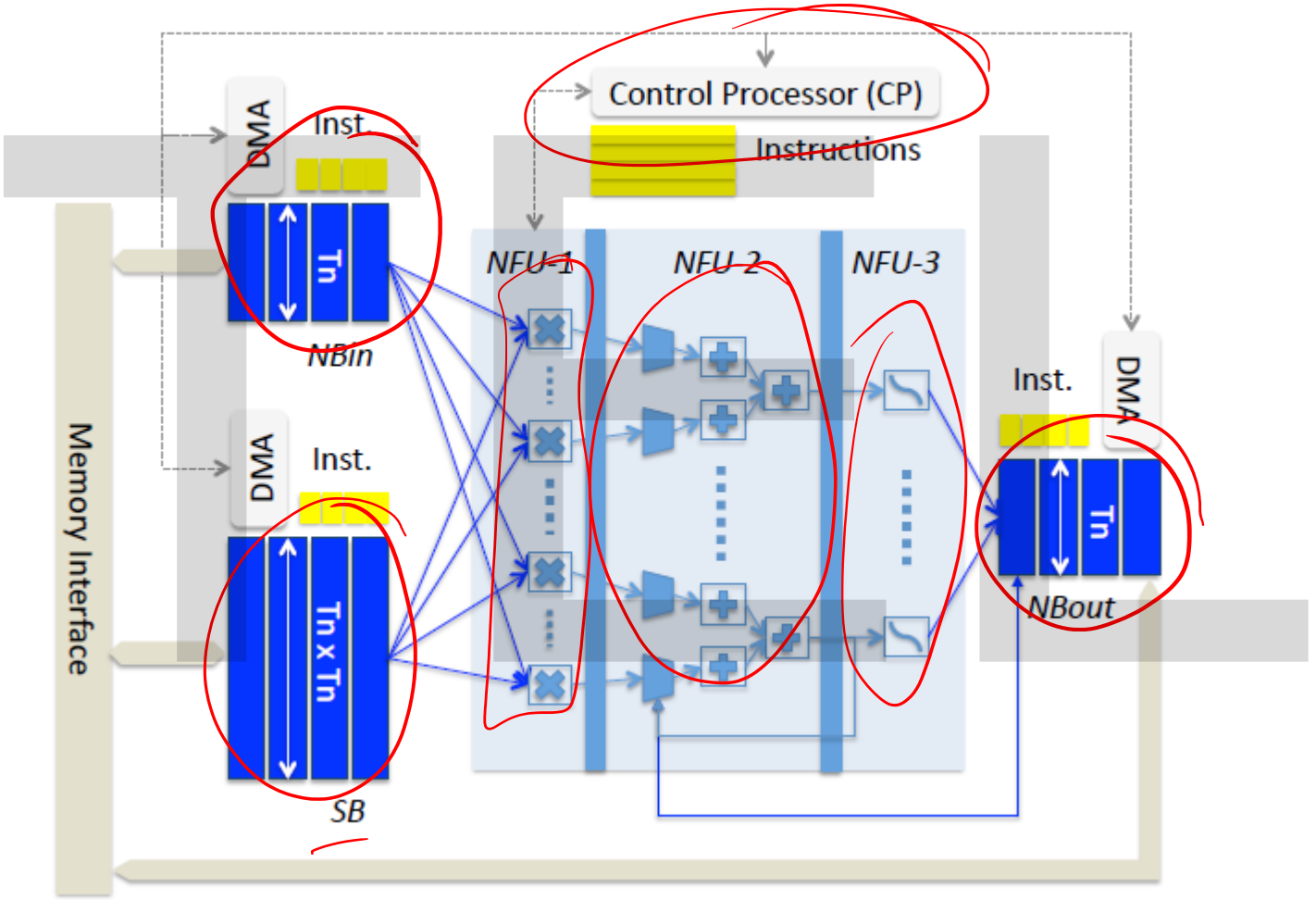
Cost of different data flow

	OS	WS	IS	Min
MACs	E*R	E*R	E*R	E*R
Weight Reads	E*R	R	E*R	R
Input Reads	E*R	E*R	E	E
Output Reads	0	E*R	E*R	0
Output Writes	E	E*R	E*R	E

Assume: W ~= E

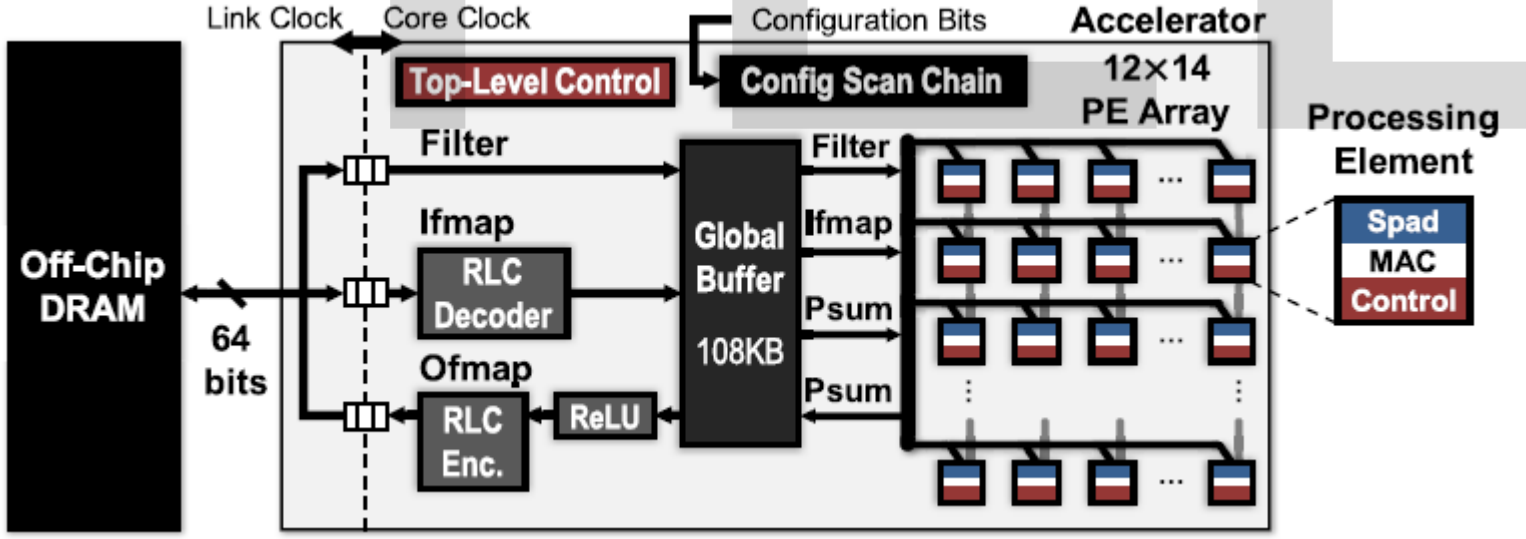
Accelerator Architecture in DianNao

- **Storage**
  - NBin: To store input activations
  - SB: To store synapses
  - NBout: To store outputs
- **Neural functional unit (NFU)**
  - NFU-1: multiplier
  - NFU-2: adder tree
  - NFU-3: non linear
- **Control instructions**



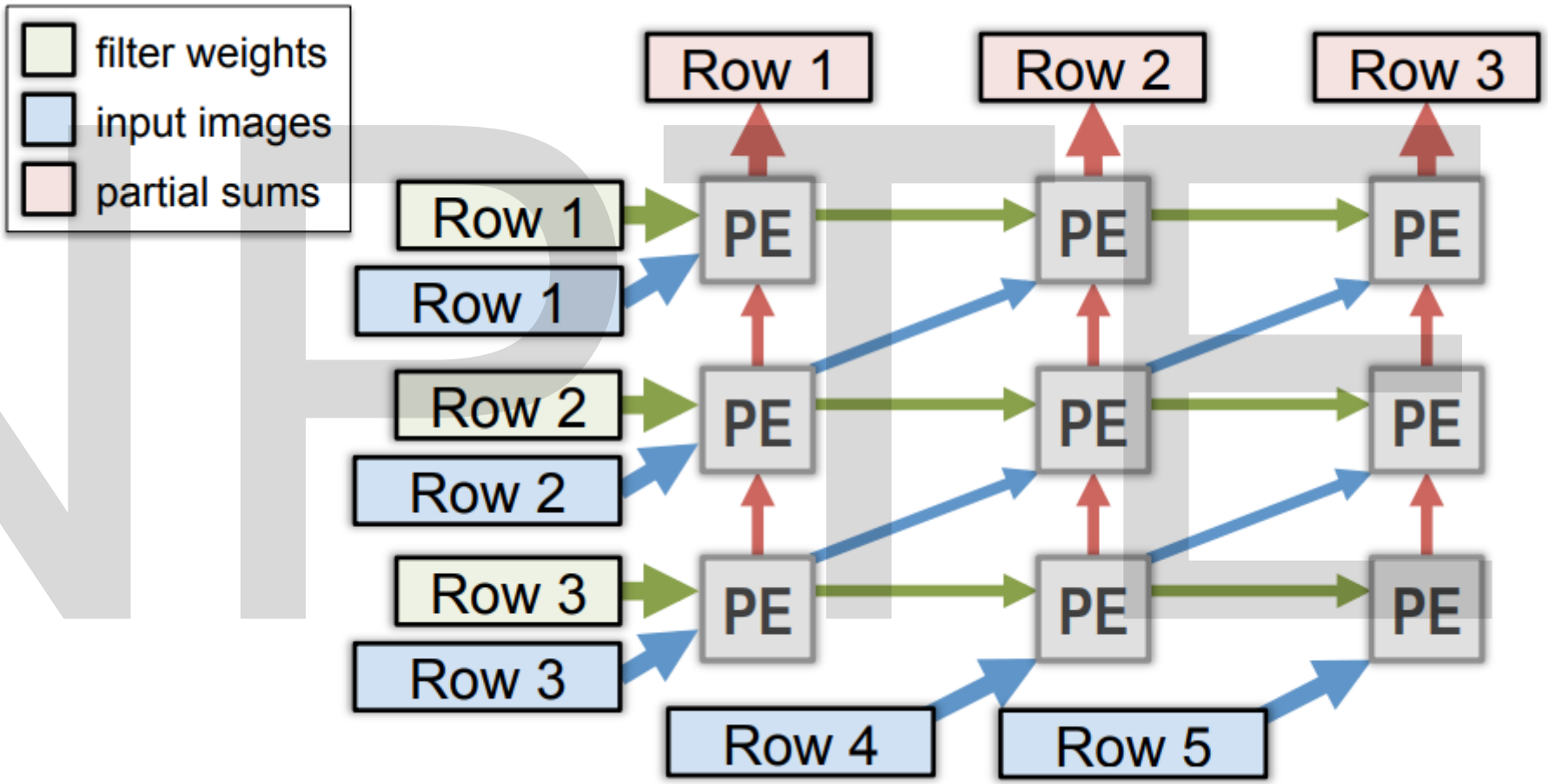
Eyeriss

- **Eyeriss: An Energy-efficient Reconfigurable Accelerator for Deep Convolutional Neural Networks**
- **What is new?**
  - New dataflow
  - New architecture
  - Encoding and decoding technique
  - Fabricated chip (a co-processor)



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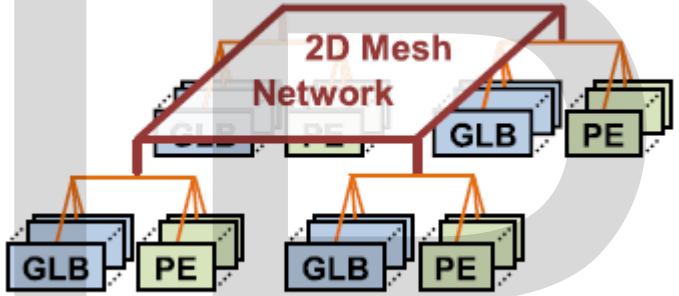
Row Stationary Dataflow (3)

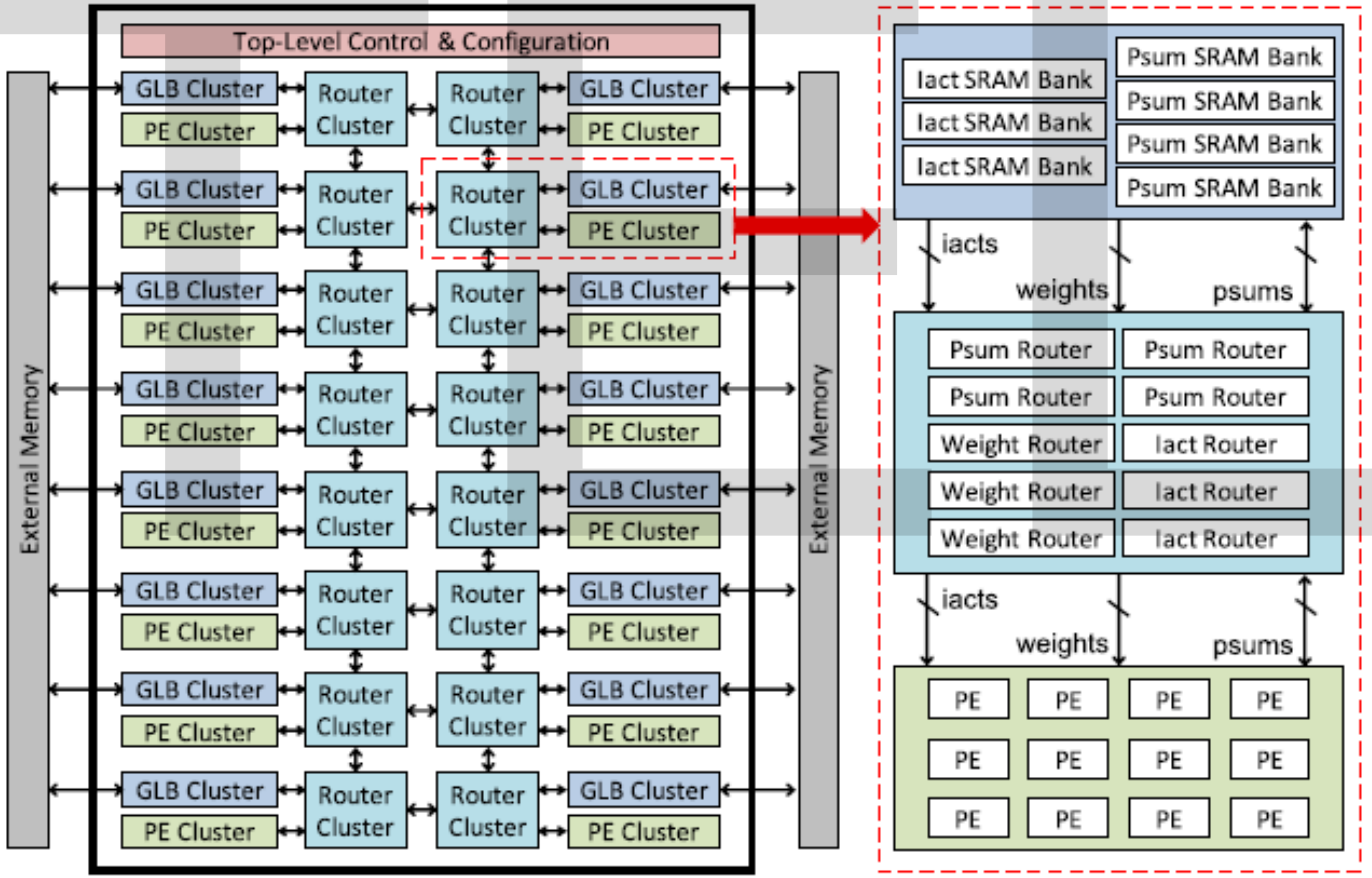


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Eyeriss-V2

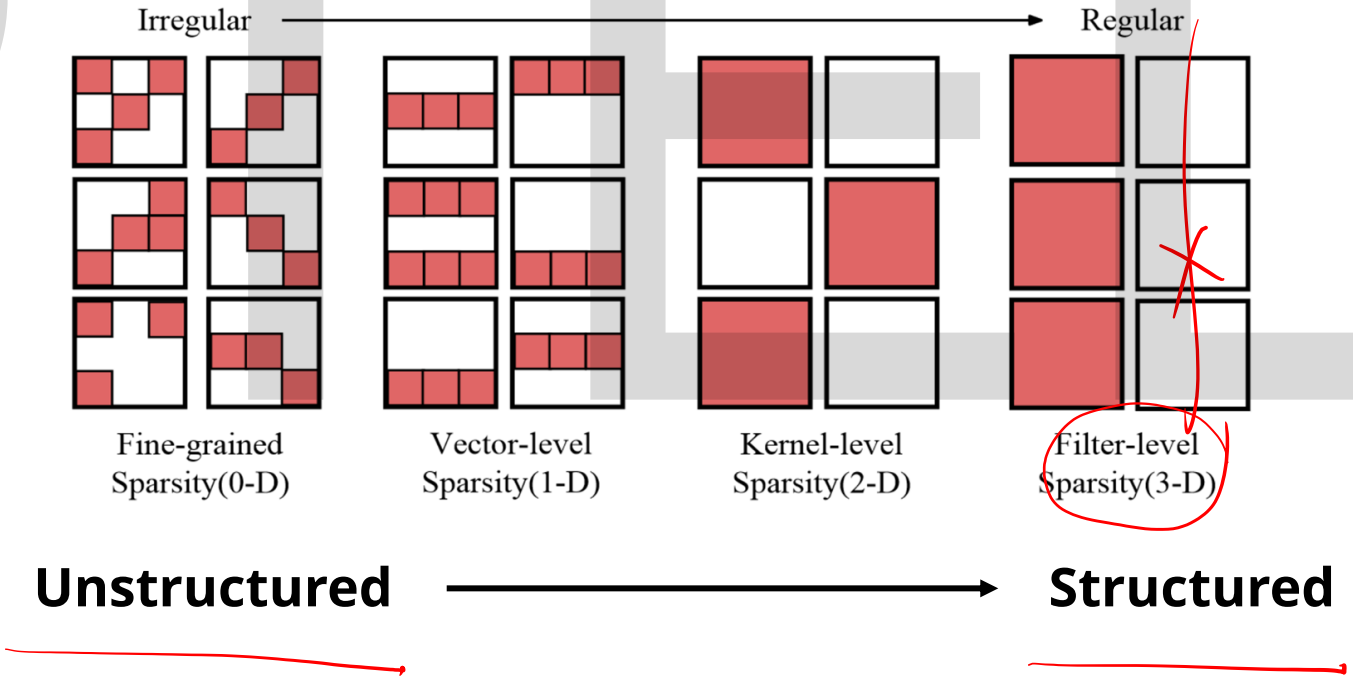
- **Distribute the communication**





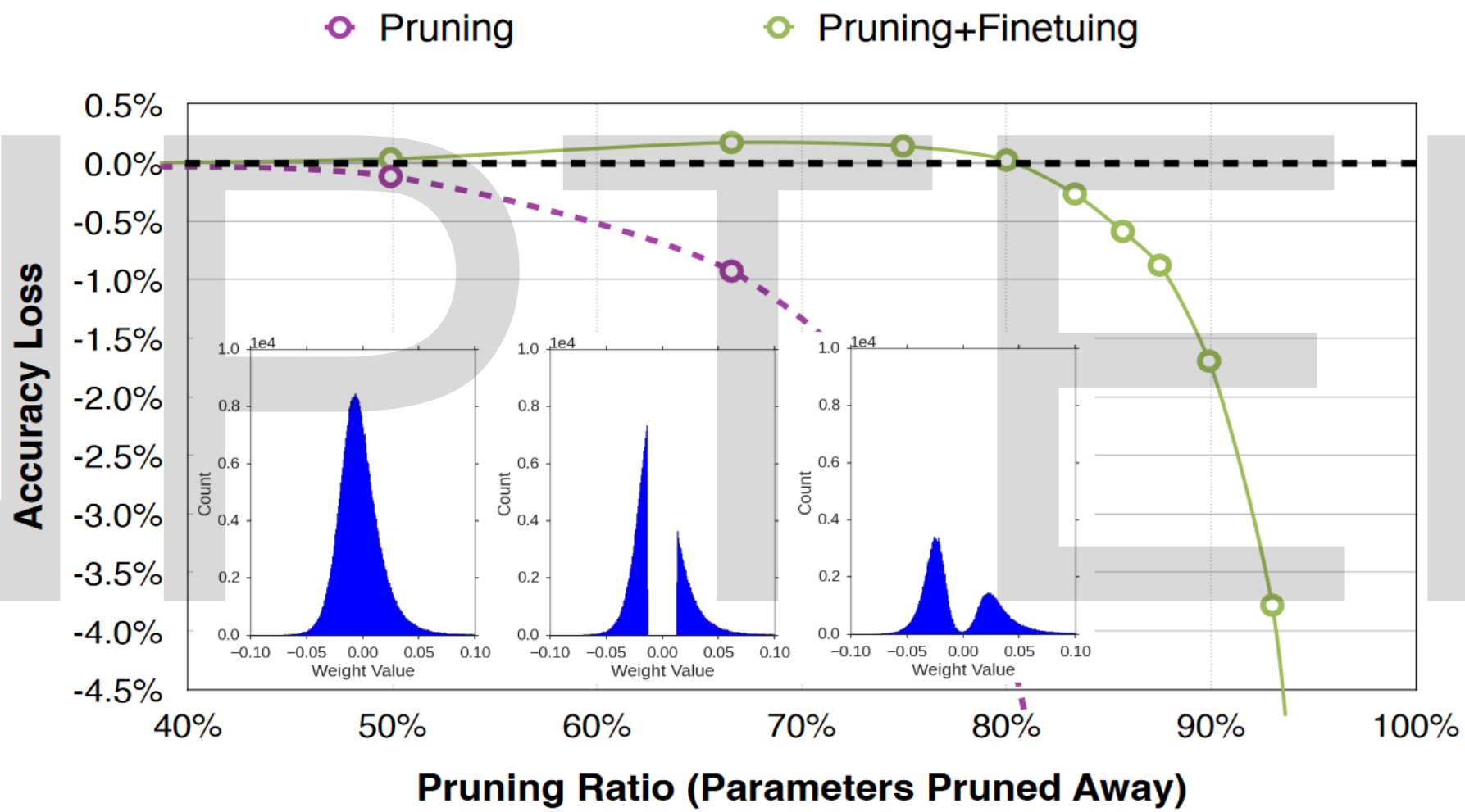
Pruning Granularity in DNNs

- **Fine Grained: Individual Weights**
- **Vector Level**
  - Sub-kernel vectors of size W
- **Kernel Level**
  - Kernels of shape H x W
- **Filter Level**
  - Filter of shape C x H x W



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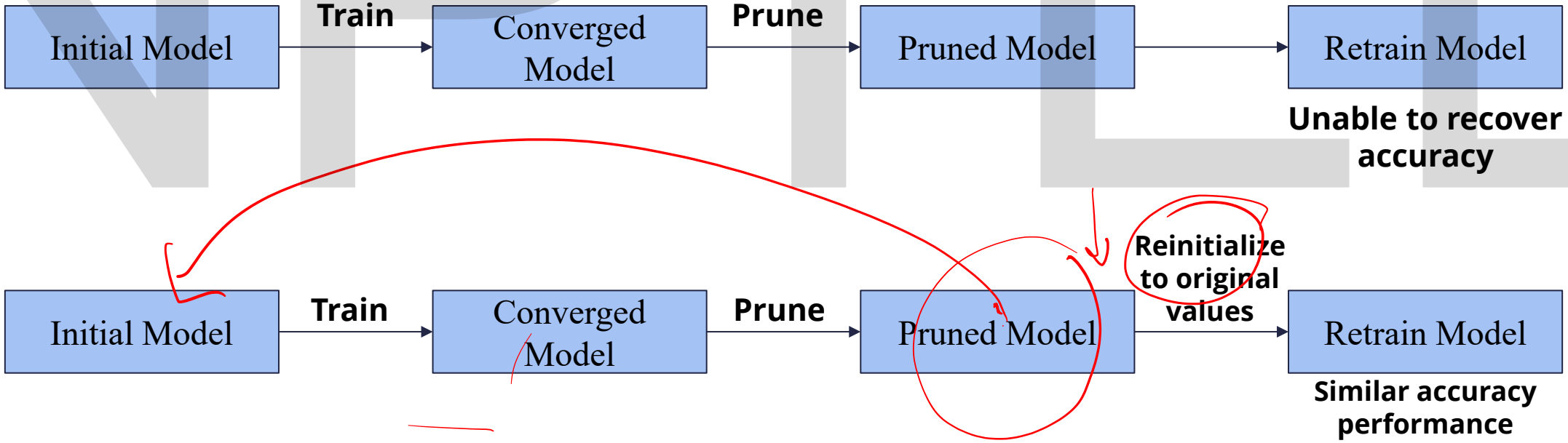
Results of Pruning and Fine tuning



Ref :Learning Both Weights and Connections for Efficient Neural Network [Han et al., NeurIPS 2015]

The Lottery Ticket Hypothesis (LTH)

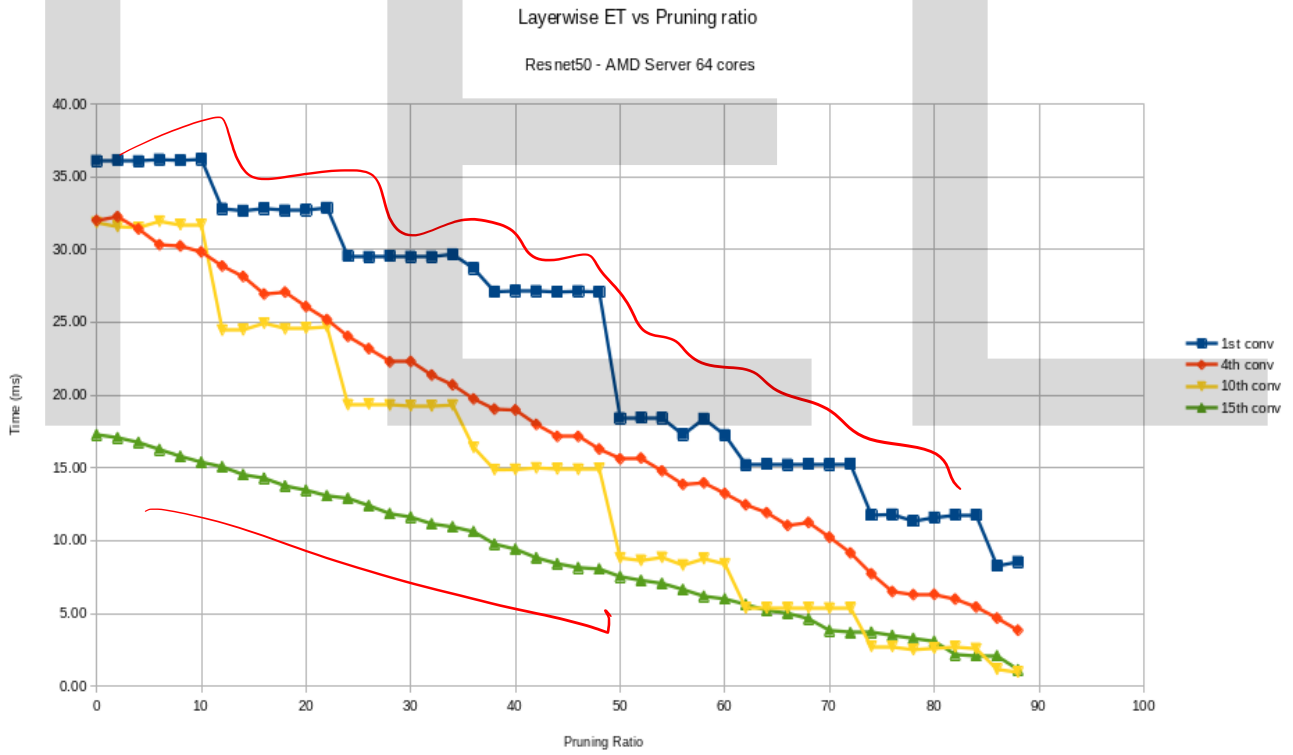
- **LTH asks the question whether we can train a sparse network which is a subset of some dense network, from scratch, and achieve the same level of accuracy**
- **But how do we identify the sparse subnetwork?**
  - Iterative magnitude pruning is one such way



Ref: The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks [Frankle et al., ICLR 2019]

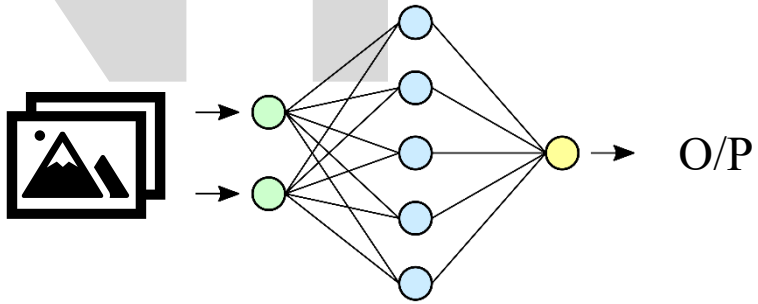
How do Execution Time Change with Pruning?

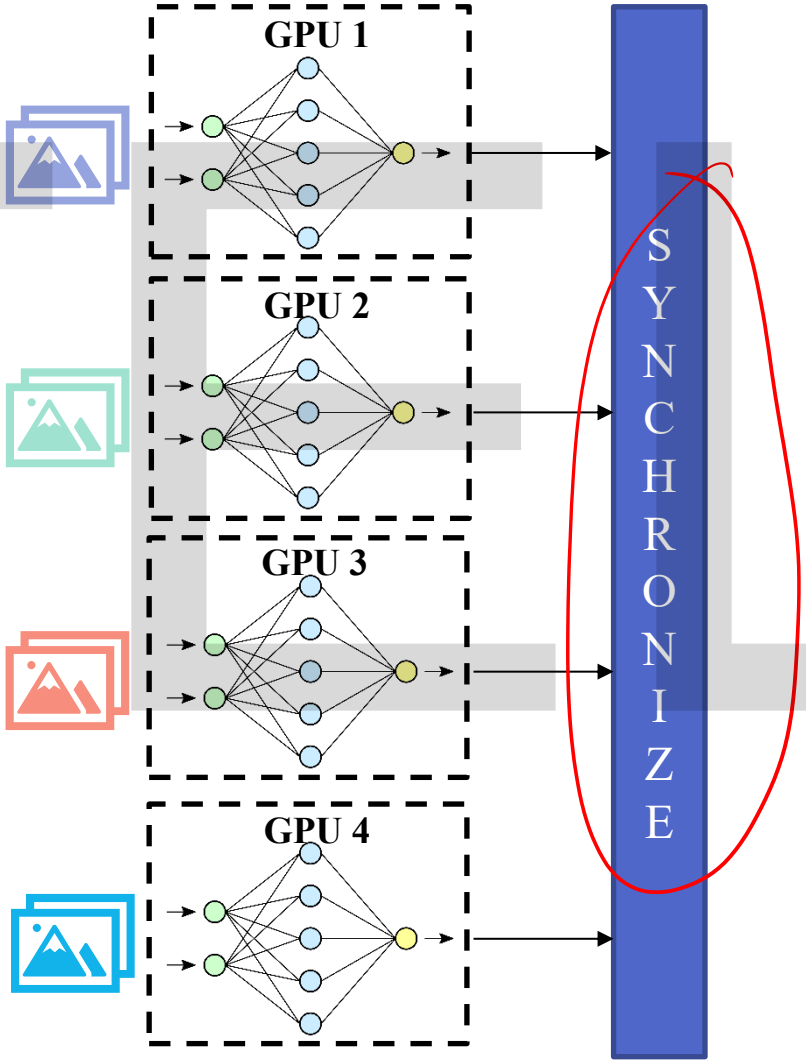
- As we prune a layer, we are reducing the number of parameters, this should result in a gradual decrease in its Execution time.
- But is this decrease in Execution time linear?
- It is observed that even for uniformly increasing pruning ratio, the execution time does not decrease linearly but in a staircase pattern



Data Parallel Training

- Replicate the model across all GPUs
- Shard (break) the inputs
- Forward and backward propagation on same model but with different data
- Average parameters (weights/bias), updates (gradients) at every iteration

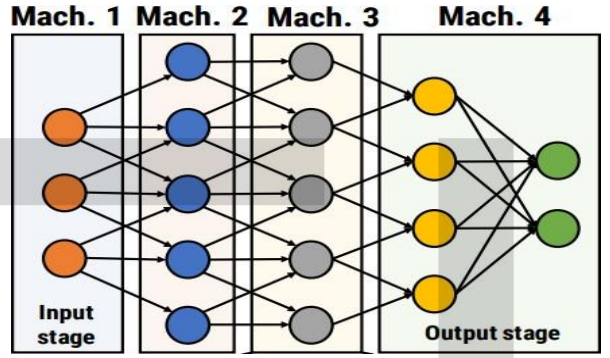


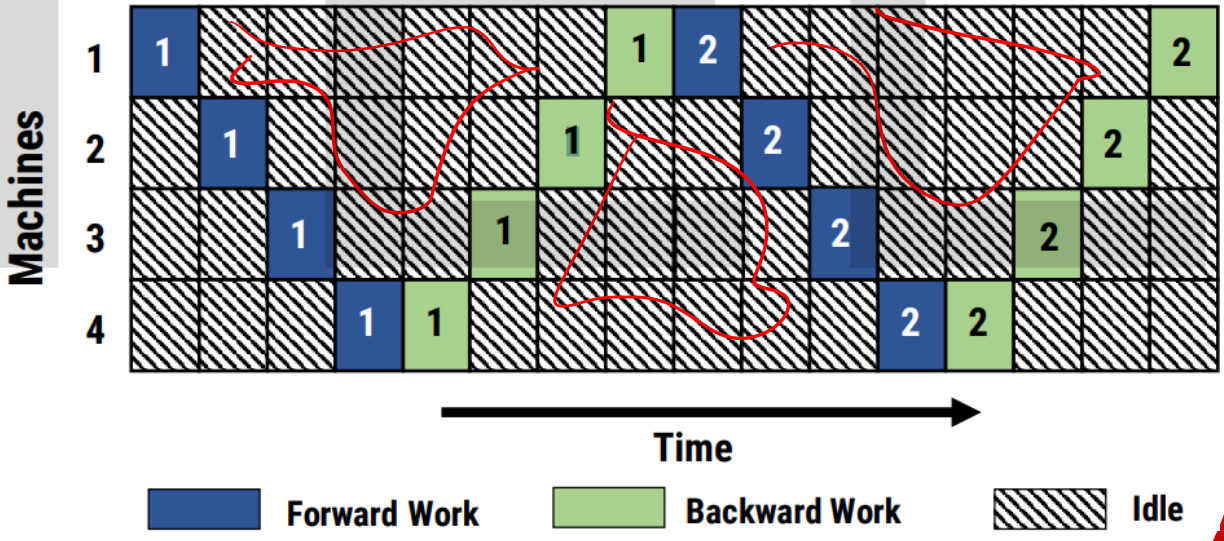


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Model Parallelism

- **Advantages**
  - Increased total memory capacity -> model size
  - Point-to-point communication -> less expensive
  - Small minibatch: better statistical efficiency
- **Disadvantages**
  - Underutilization
  - Cannot mask communication
- **How to partition the model over the workers?**

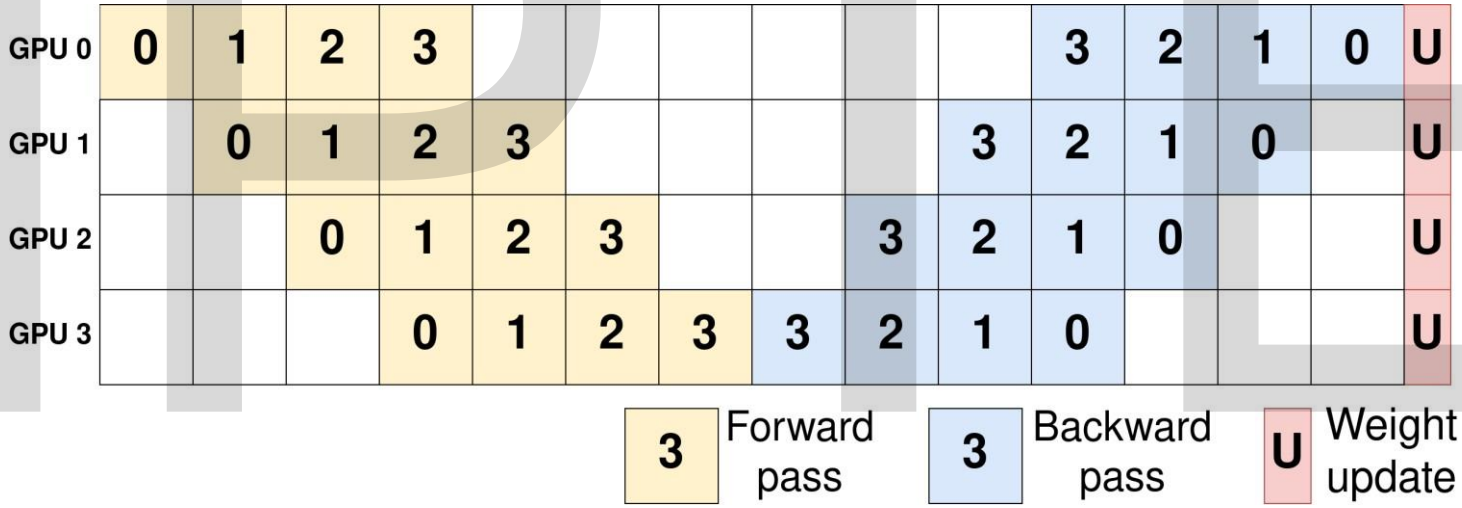




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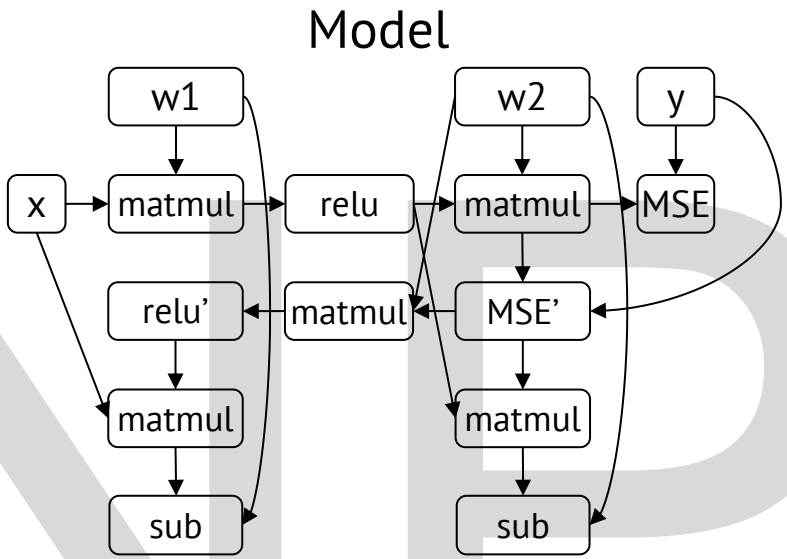
Pipeline Parallelism

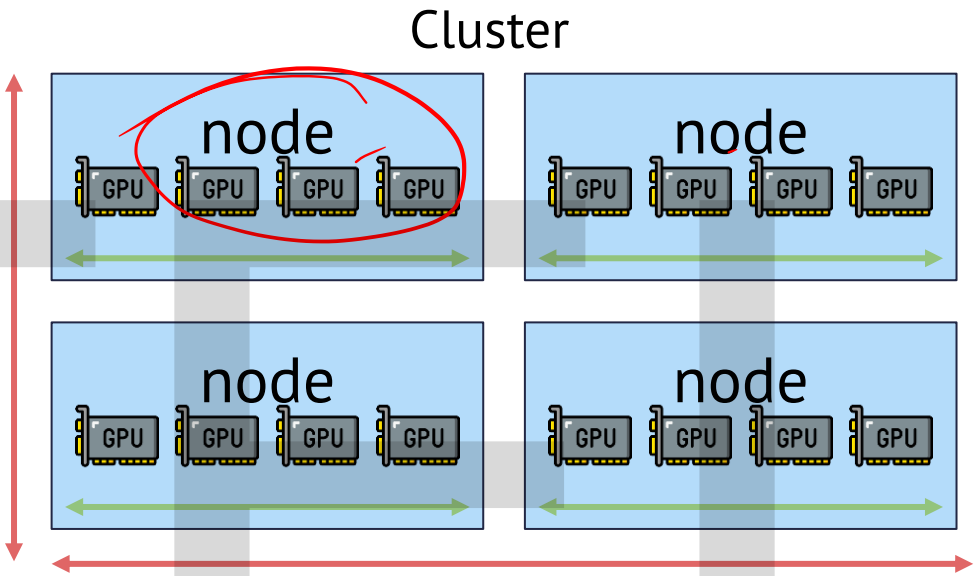
- **Similar to model parallelism, but use multiple batches in a pipelined fashion**

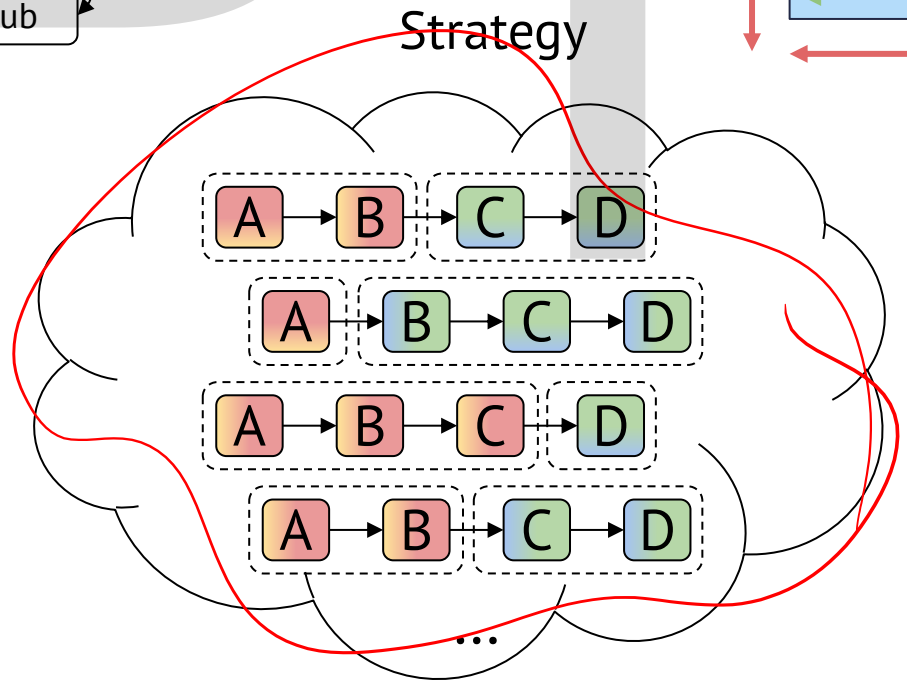


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Auto-parallelization: Problem

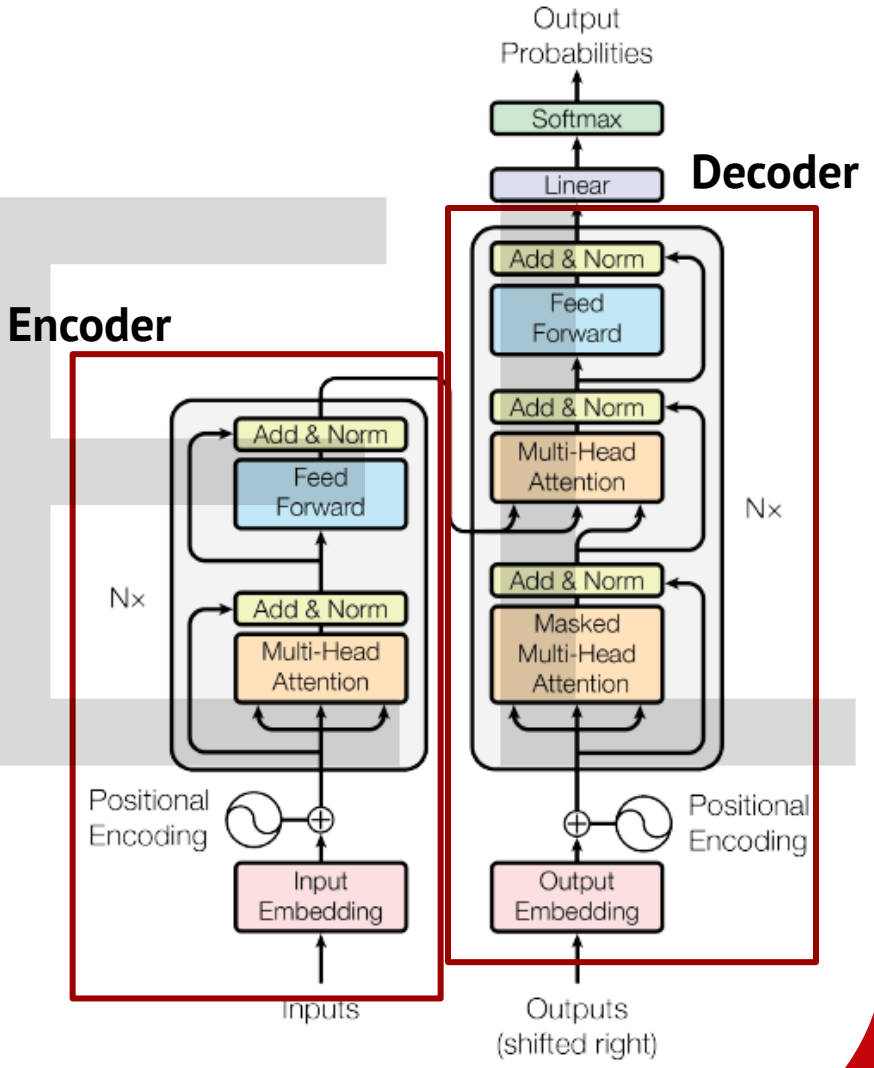






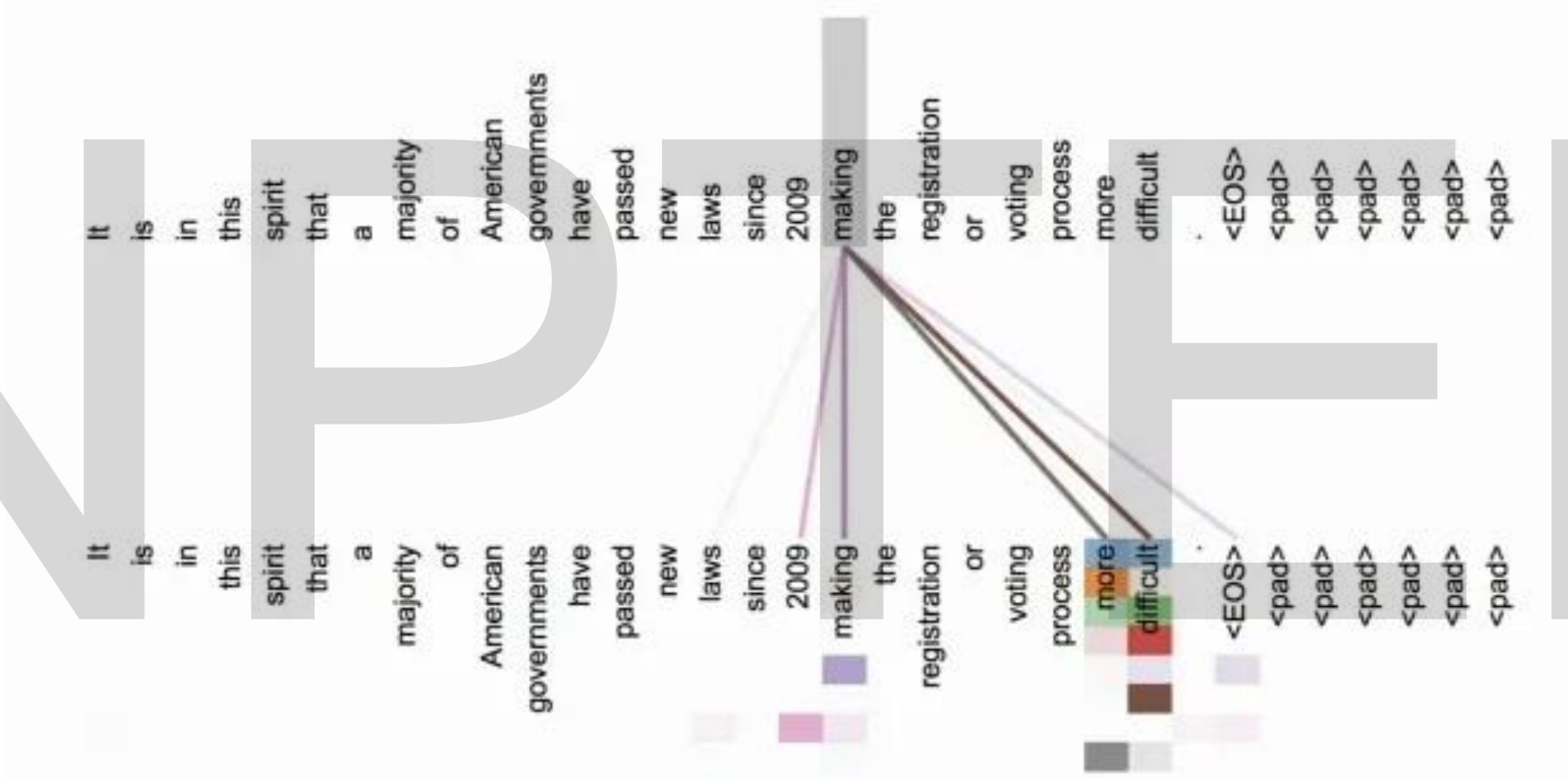
Attention is All you Need [1]

- **Based on Encoders and Decoders**
  - **They are connected**
- **Input to Encoder is a sequence of words (sentence)**
- **Input to Decoder**
  - **Outputs (shifted right)**
  - **Output probabilities**
  - **Input from Encoder**



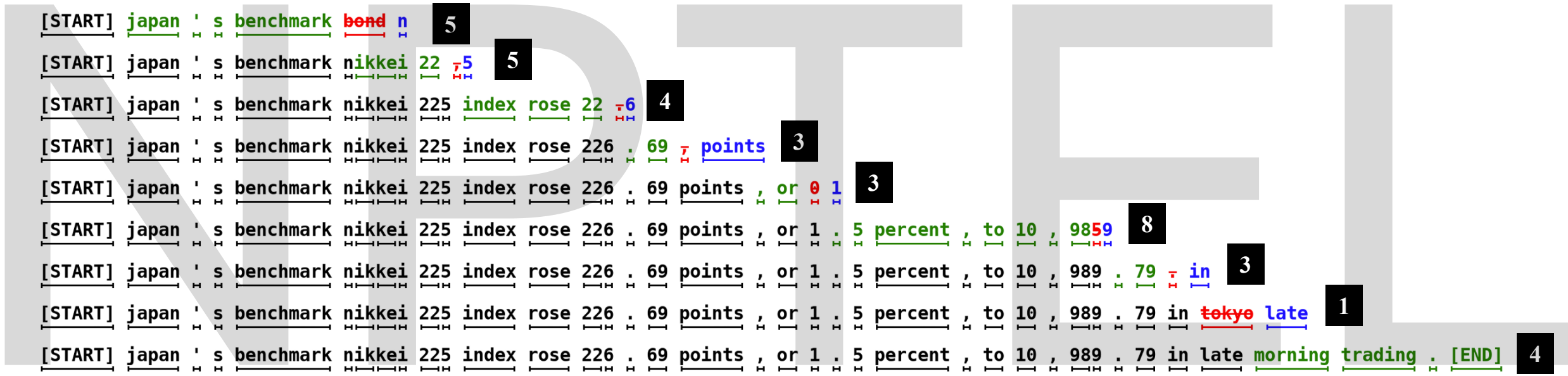
[1] Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Visualizing Attentions



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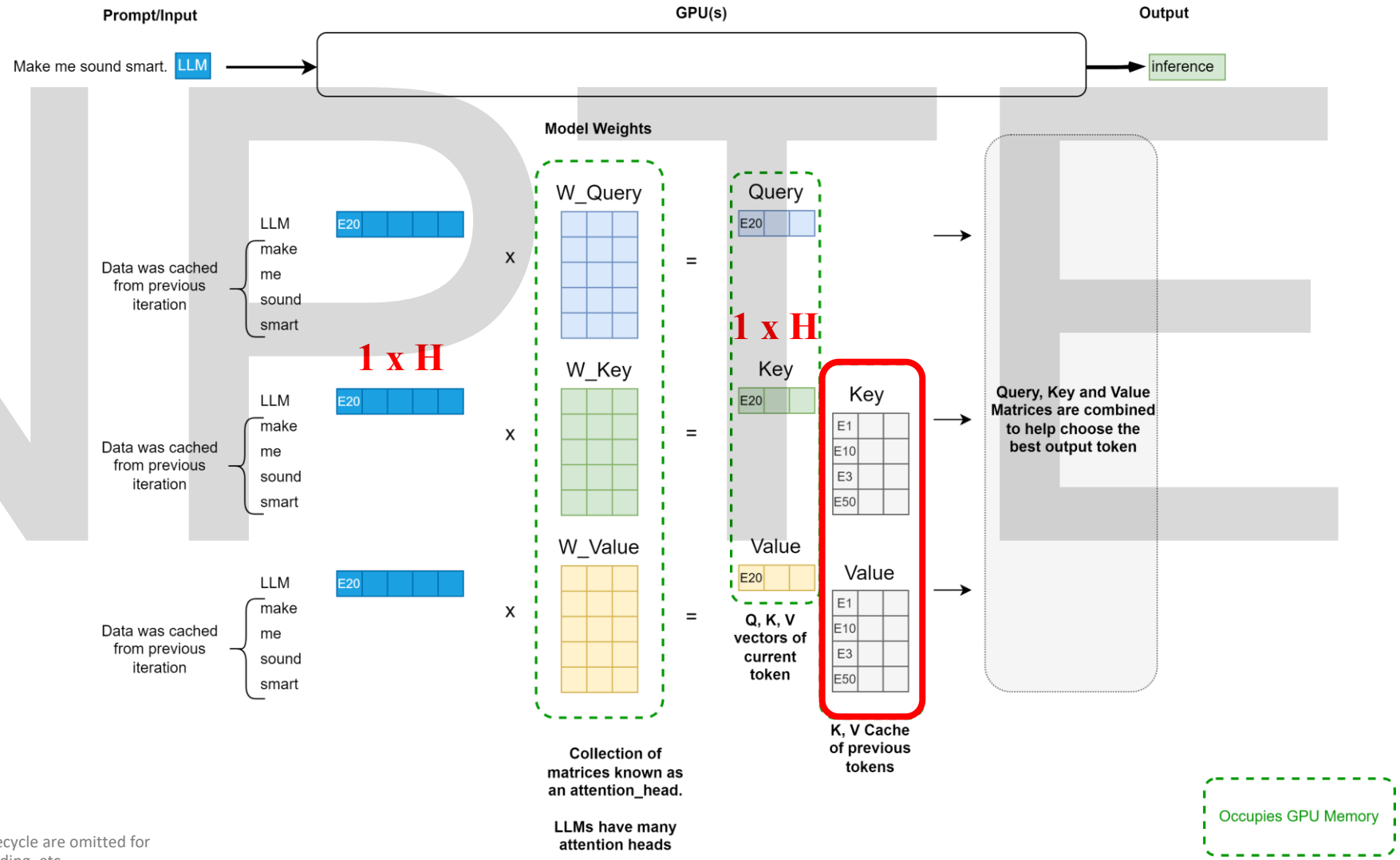
Speculative Decoding: Example



37 forward pass vs 9 forward pass!

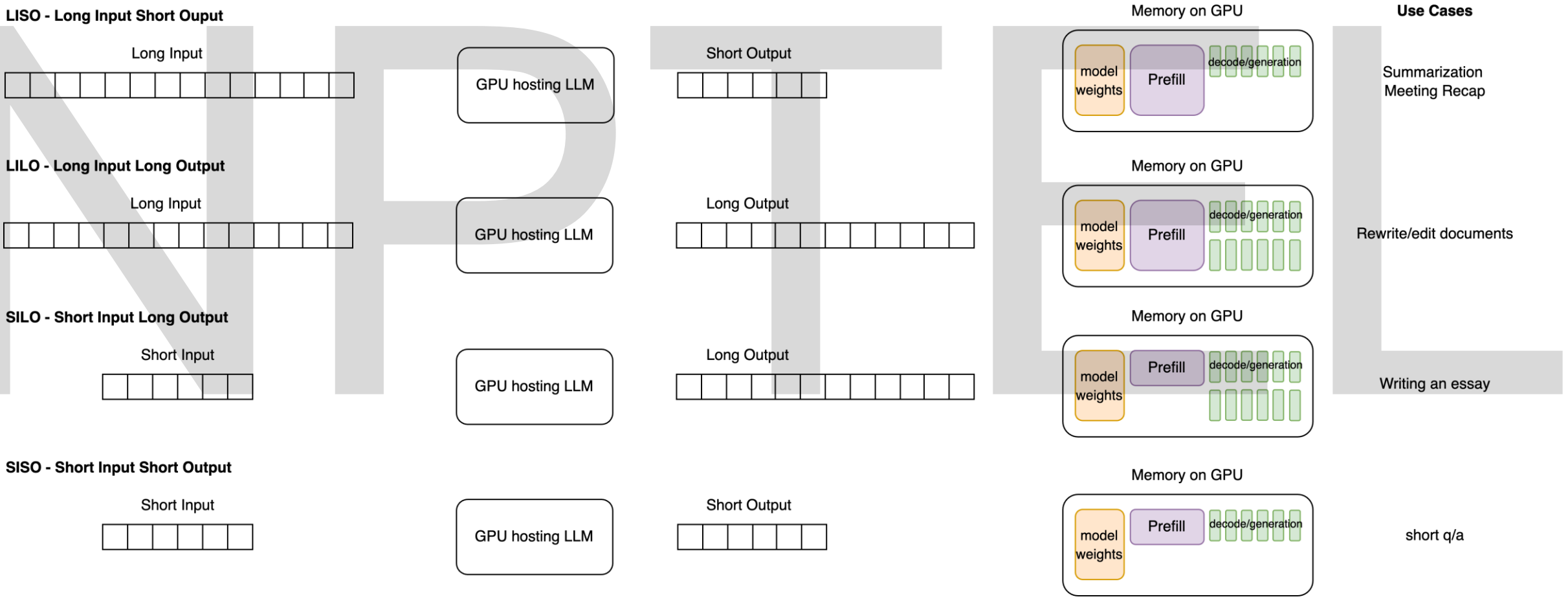
What about the forward passes through the draft model?

KV cache to the rescue!



Many details of the model inference lifecycle are omitted for brevity such as mlps, layer-norms, decoding, etc

Why KV cache memory management?



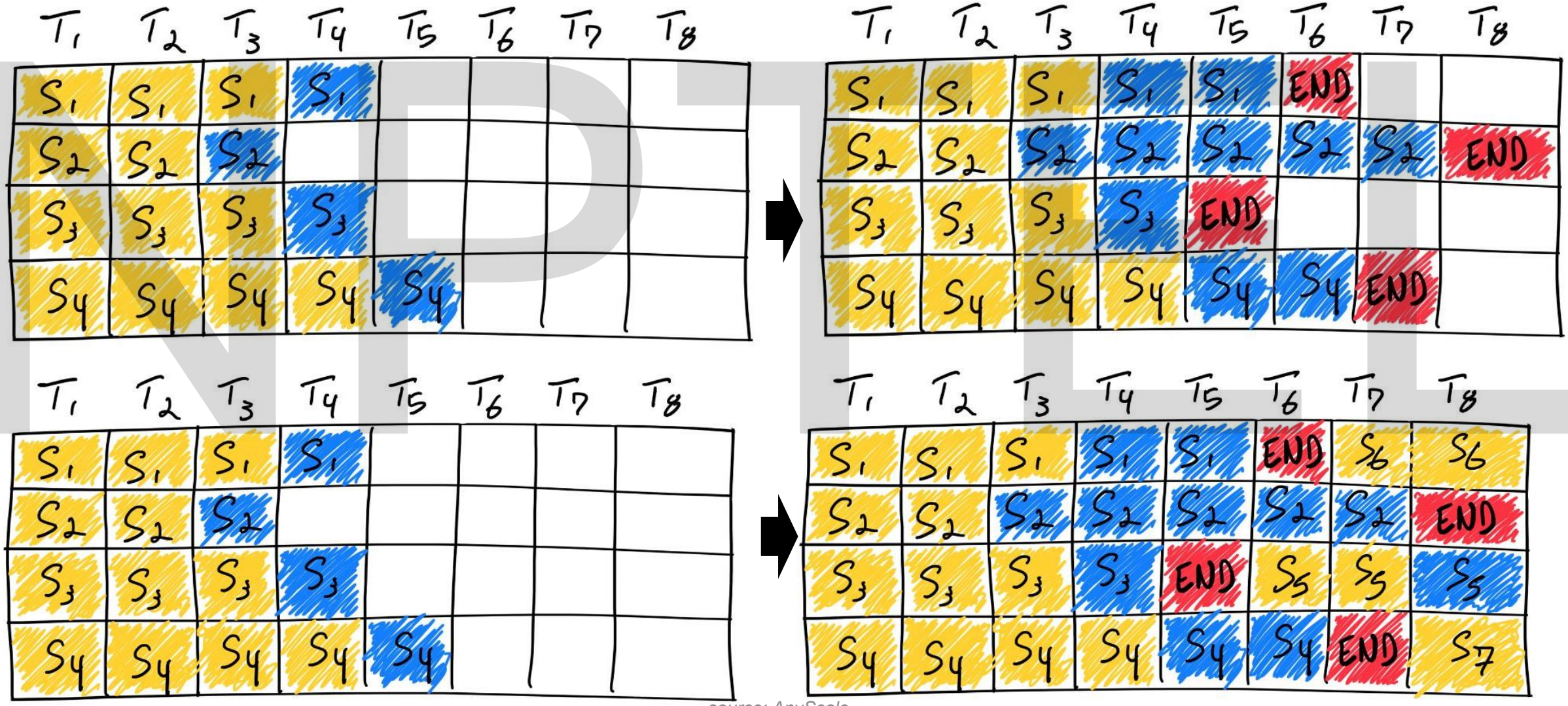
LLM Serving

Top: static batching
Bottom: continuous batching

Legend:

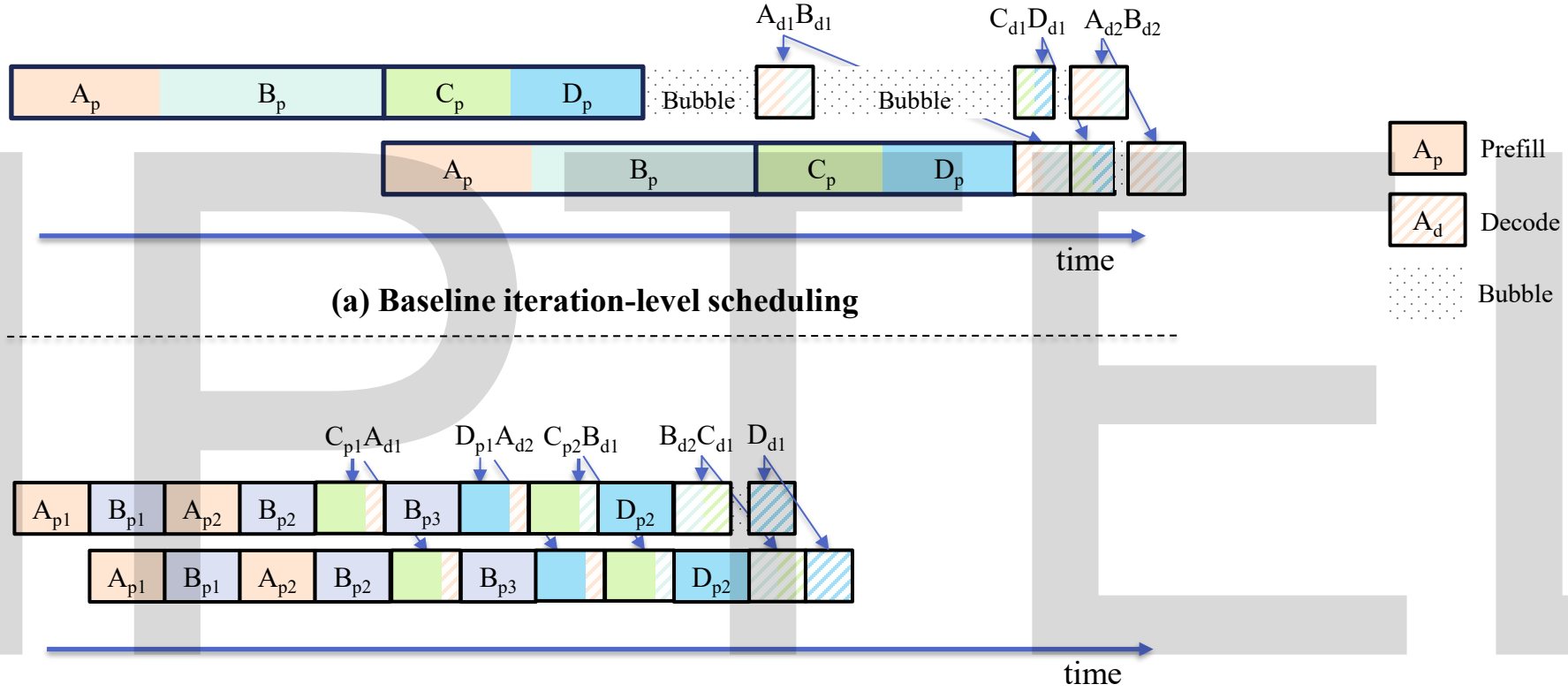
- Yellow: prompt token
- Blue: generated token
- Red: end-of-sequence token

Orca [OSDI 22]



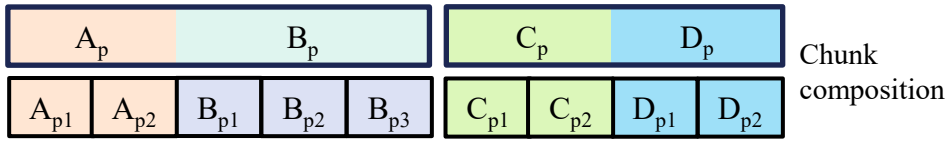
source: AnyScale

SARATHI: Chunked Prefill

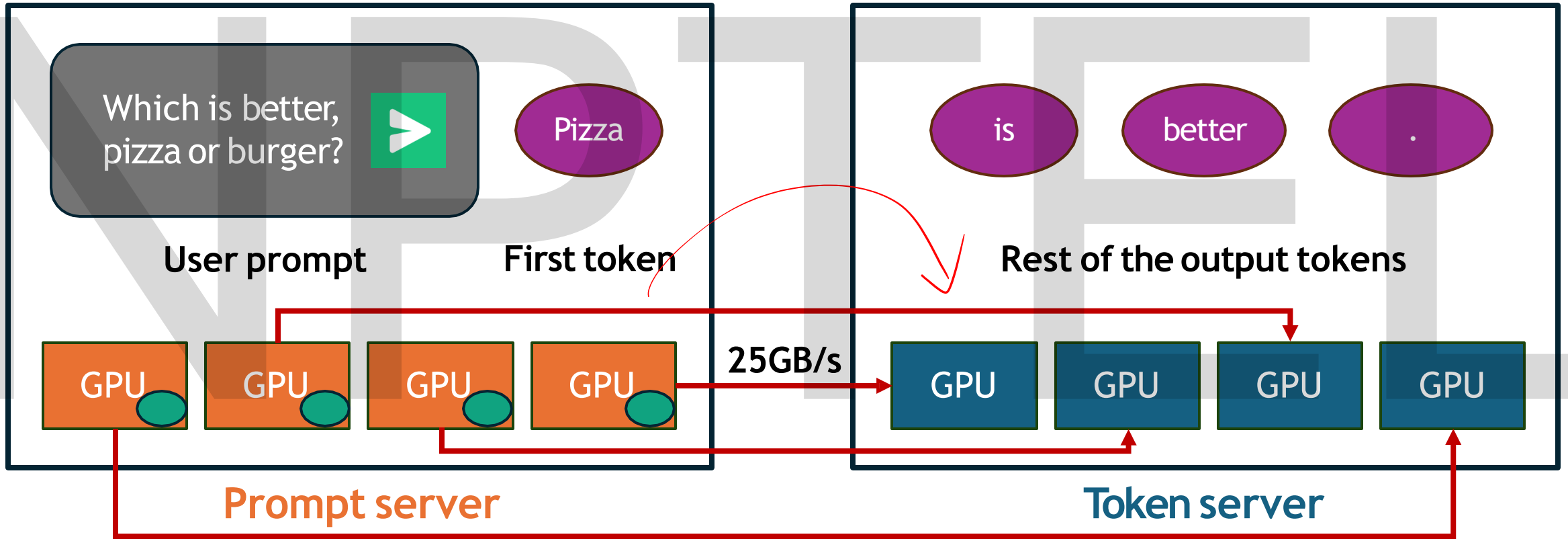


(a) Baseline iteration-level scheduling

(b) SARATHI : Chunked prefills with decode-maximal batching



Segregating Prefill and Decode Server



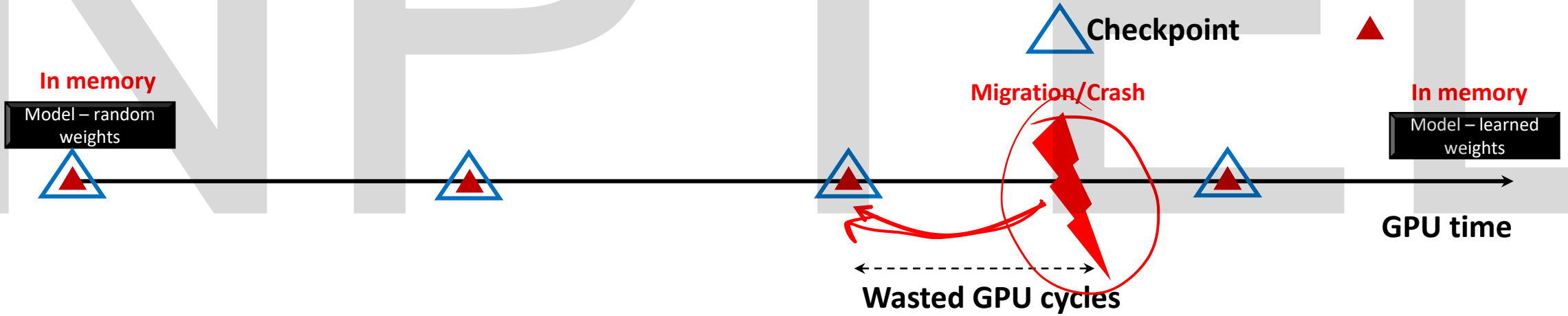
: 1/4th of the KV cache

Total bandwidth: 100GB/s

Assuming 4-way tensor parallelism

Fault Tolerance in ML Training

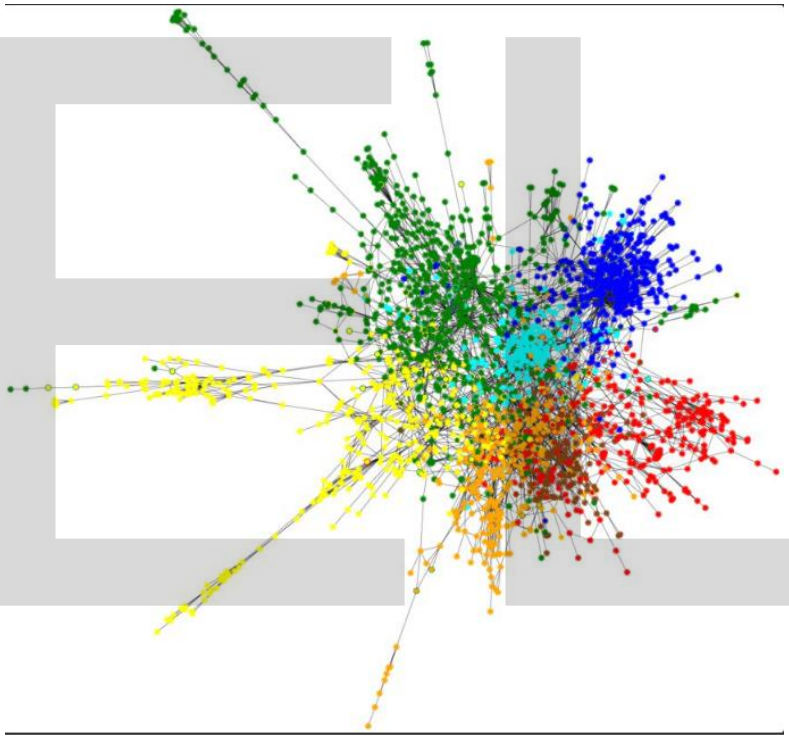
- Learned model parameters are written to persistent storage every so often during training for fault-tolerance
  - The VMs may migrate, expire or crash, jobs may migrate (e.g., shared GPU clusters)



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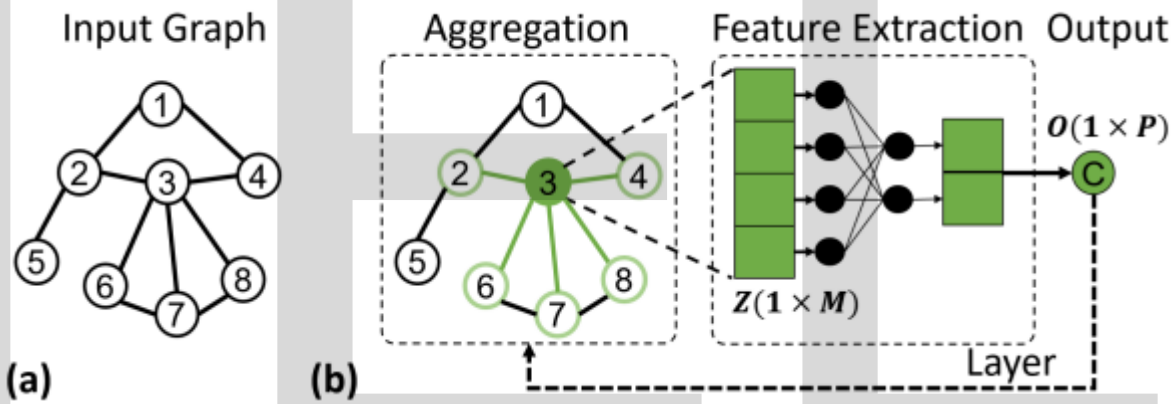
Representation of Cora Dataset

- 2708 scientific publications
- Classified into one of seven classes
  - Each colour denotes a class
- The network consists of 5429 links (citations)
- Each publication is represented by a 0/1-valued word vector indicating absence/presence of the corresponding word from the dictionary (features)
- The dictionary consists of 1433 unique words



Classification of Graph Nodes through GNN

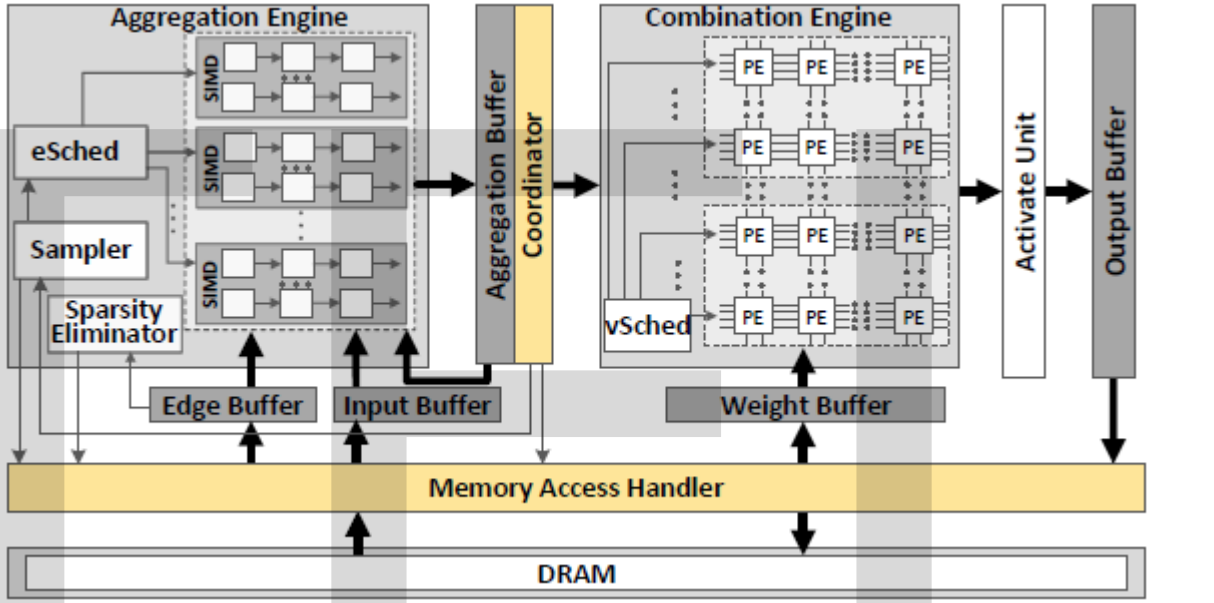
- **Two phases**
  - Aggregation
  - Combination/Feature Extraction
- **Aggregate function aggregates multiple feature vectors from neighbours to form one single feature vector**
  - Different GNNs use different function
- **Combination function transforms the feature vector of each vertex to another feature vector using a MLP**
- **MLP parameters are shared between vertices**



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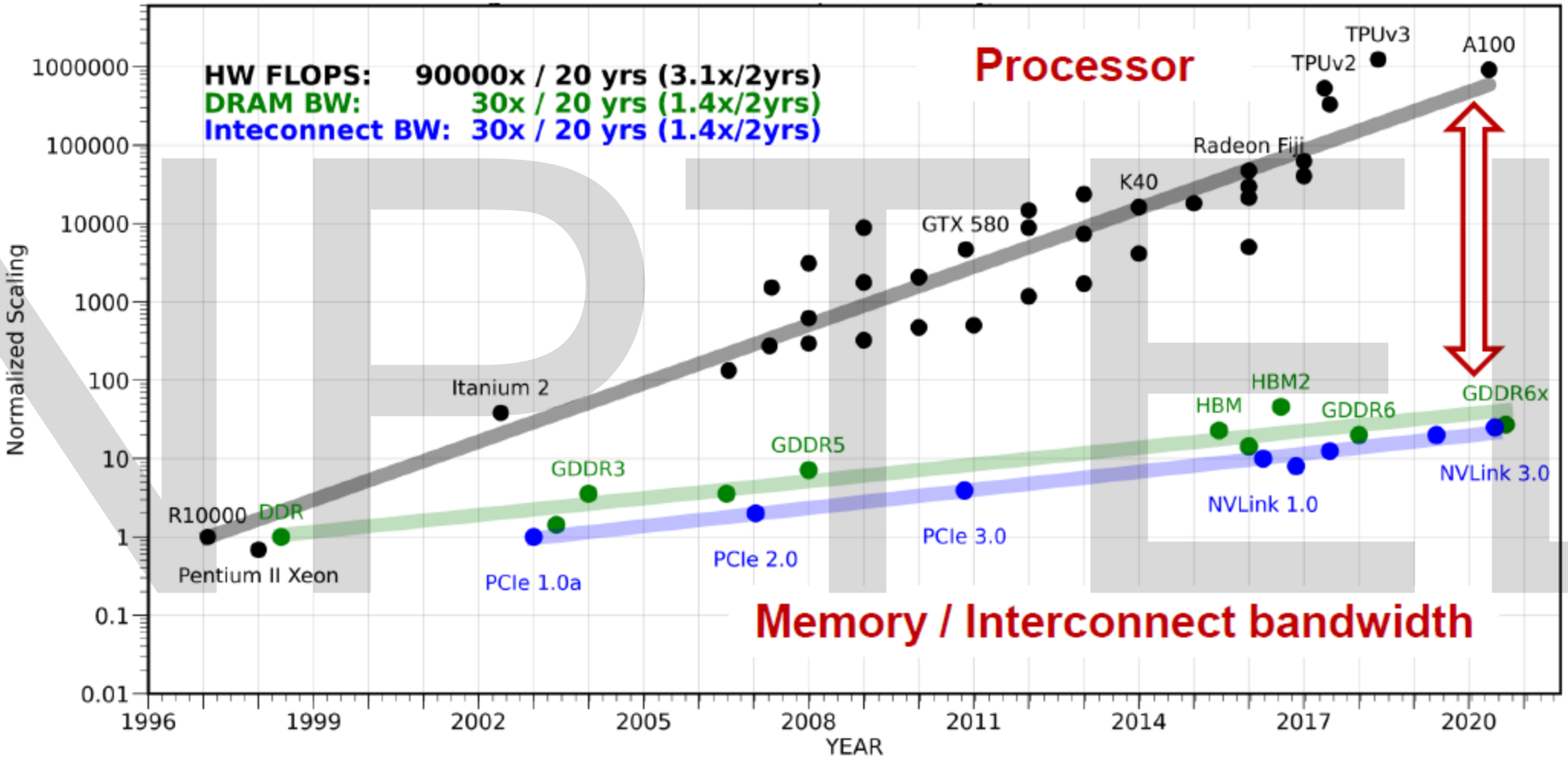
HyGCN Architecture [1]

- HyGCN
  - A hybrid architecture
  - Two engines: Aggregation and Combination
  - Memory access Handler
  - A communication interface (coordinator)



[1] Yan, Mingyu, et al. "Hygcn: A gcn accelerator with hybrid architecture." 2020 IEEE International Symposium on High Performance Computer Architecture (HPCA). IEEE, 2020.

Need for In-Memory Computation



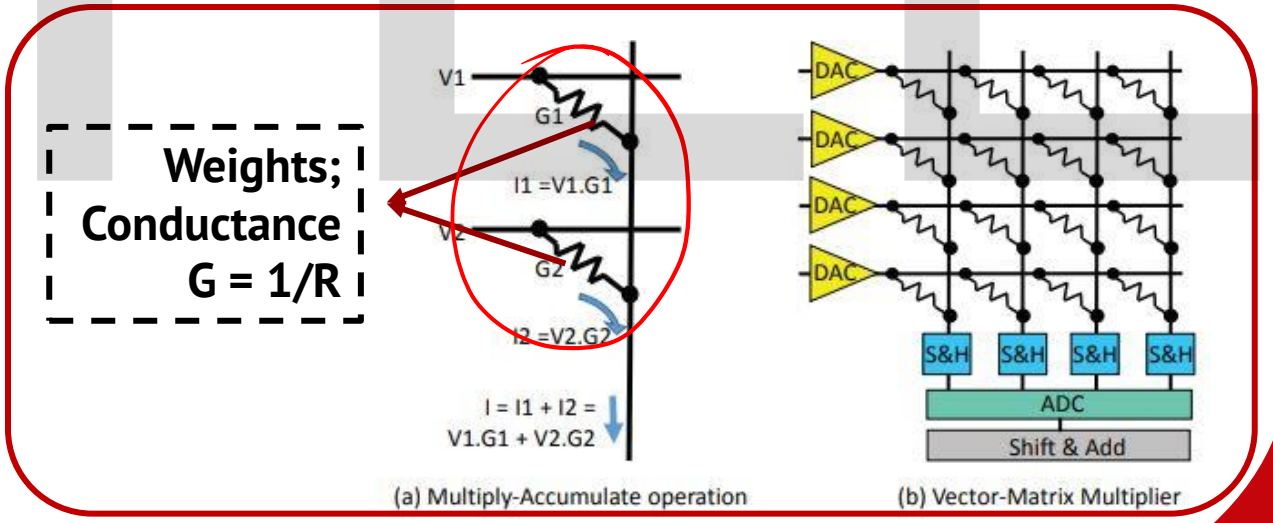
[1] Work by A. Gholami, 2020; Hynix; Chipworks; Created by Prof. Yu Cao, ASU.

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In-memory Computing (IMC) [1]

- **Analog computing: resistive (current) or capacitive (charge)**
- **Example: Resistive Random Access Memory (ReRAM or RRAM)**
  - A nonvolatile storage
  - Operates by changing the resistance of a special dielectric material
  - A family of Memristor
- **Advantage: No data movement between off chip memory**

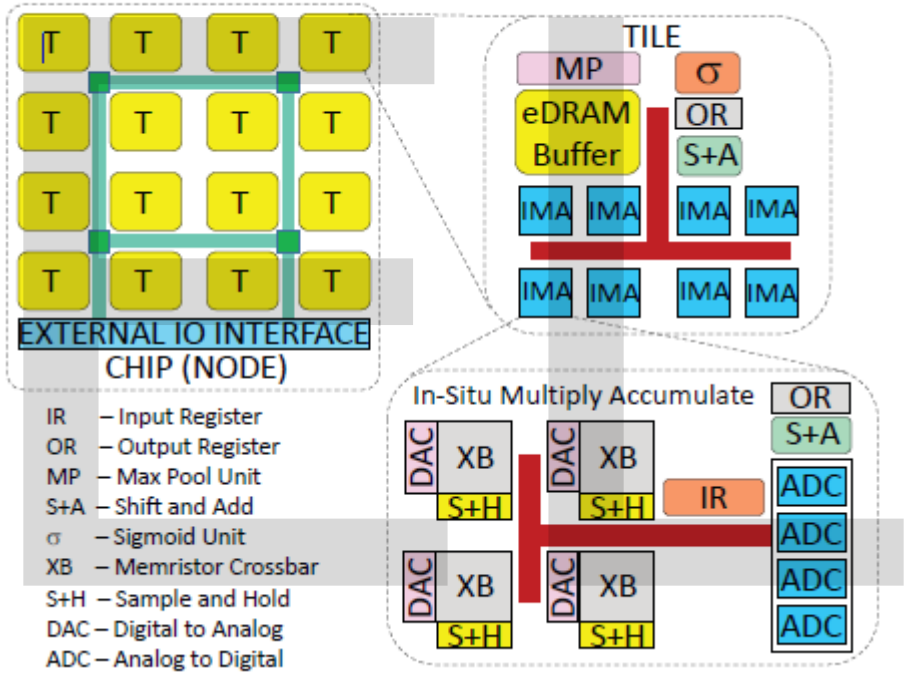
[1] Shafiee, Ali, et al. "ISAAC: A convolutional neural network accelerator with in-situ analog arithmetic in crossbars." ***ACM SIGARCH Computer Architecture News*** **44.3 (2016): 14-26.**



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Overview of ISAAC

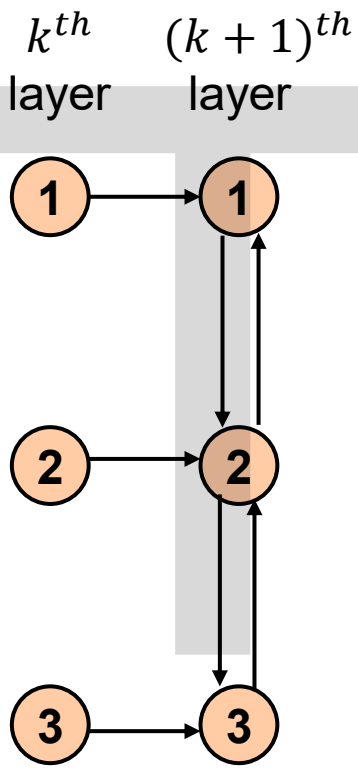
- **The ISAAC chip consists of number of tiles**
  - Connected with concentrated-mesh
  - eDRAM buffers to store input values
  - In-situ multiply-accumulate (IMA) units
  - Output registers to aggregate results
- **IMA**
  - Crossbars (XB)
  - DAC, ADC
  - Sample and hold
- **Only inference**



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Solution: Custom Interconnect [1]

- Two consecutive layers consists of equal number of routers
  - Three routers in this case
- Each router can send/receive only one packet at one cycle
- In one round of communication each routers in k^{th} layer send a packet to each router in $(k+1)^{th}$ layer
- One transaction is finished in 3 cycles
  - Minimum possible latency



(a)

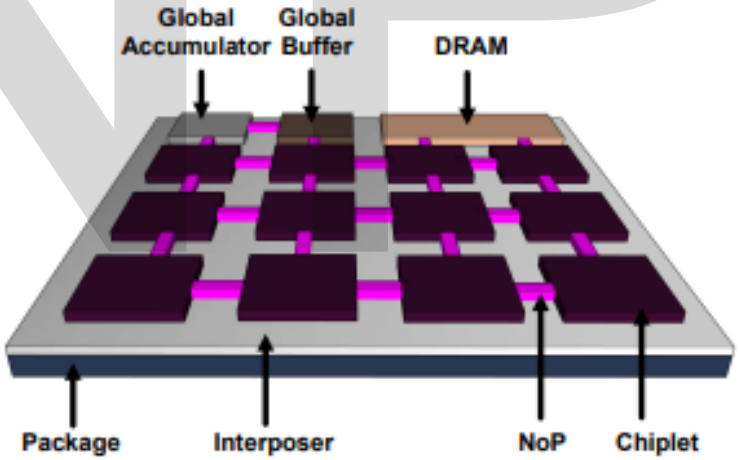
Link	Cycle	Transaction
$L_{k,1} - L_{(k+1),1}$	1	$R_{k,1} - R_{(k+1),1}$
$L_{k,2} - L_{(k+1),2}$	1	$R_{k,2} - R_{(k+1),2}$
$L_{k,3} - L_{(k+1),3}$	1	$R_{k,3} - R_{(k+1),3}$
$L_{(k+1),1} - L_{(k+1),2}$	2	$R_{k,1} - R_{(k+1),2}$
$L_{(k+1),2} - L_{(k+1),3}$	2	$R_{k,2} - R_{(k+1),3}$
$L_{(k+1),3} - L_{(k+1),2}$	2	$R_{k,3} - R_{(k+1),2}$
$L_{(k+1),2} - L_{(k+1),1}$	2	$R_{k,2} - R_{(k+1),1}$
$L_{(k+1),2} - L_{(k+1),3}$	3	$R_{k,1} - R_{(k+1),3}$
$L_{(k+1),2} - L_{(k+1),1}$	3	$R_{k,3} - R_{(k+1),1}$

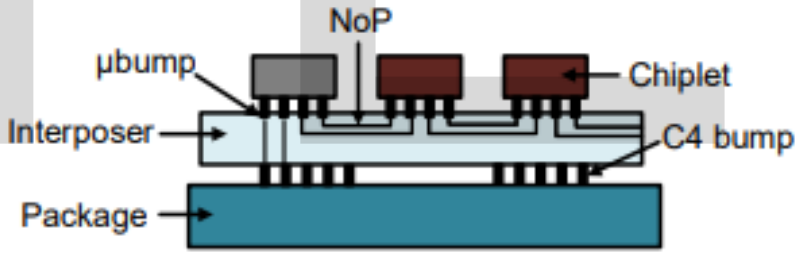
(b)

[1] Mandal, Sumit K., et al. "A latency-optimized reconfigurable NoC for in-memory acceleration of DNNs." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 10.3 (2020): 362-375.

Background (2.5D Technology)

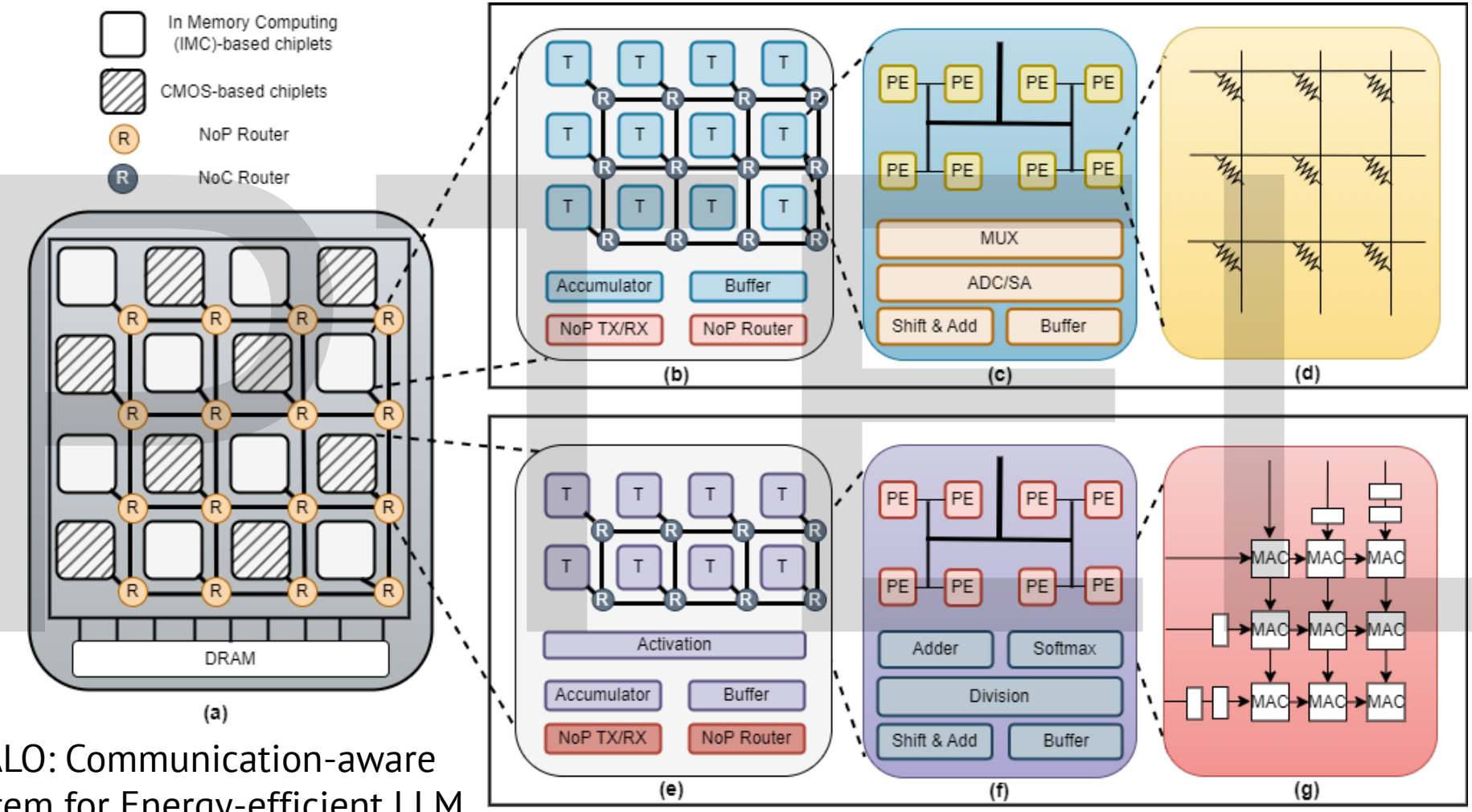
- Yield for monolithic IC is low
- 3D IC shows thermal hotspots due to increasing number of through-silicon vias (TSV)
- 2.5D technology has emerged recently which increases yield without much of





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Heterogeneous Architecture for LLMs [1]



[1] Jaiswal, Abhi, et al. "HALO: Communication-aware Heterogeneous 2.5 D System for Energy-efficient LLM Execution at Edge." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* (2024).

Performance Characterization of ML Algorithms

- Performance characterization is the process of **measuring, analyzing, and quantifying** how a **system**, device, or algorithm **behaves under various conditions**. Its primary goal is to establish baselines, identify bottlenecks, and verify whether a product meets its functional and efficiency requirements.
- Studying performance characterization is essential because it **bridges the gap between theoretical system design and real-world execution**. It empowers engineers to **identify bottlenecks, optimize resource allocation, and ensure robust, cost-effective operations** in everything from computing algorithms to physical infrastructure.

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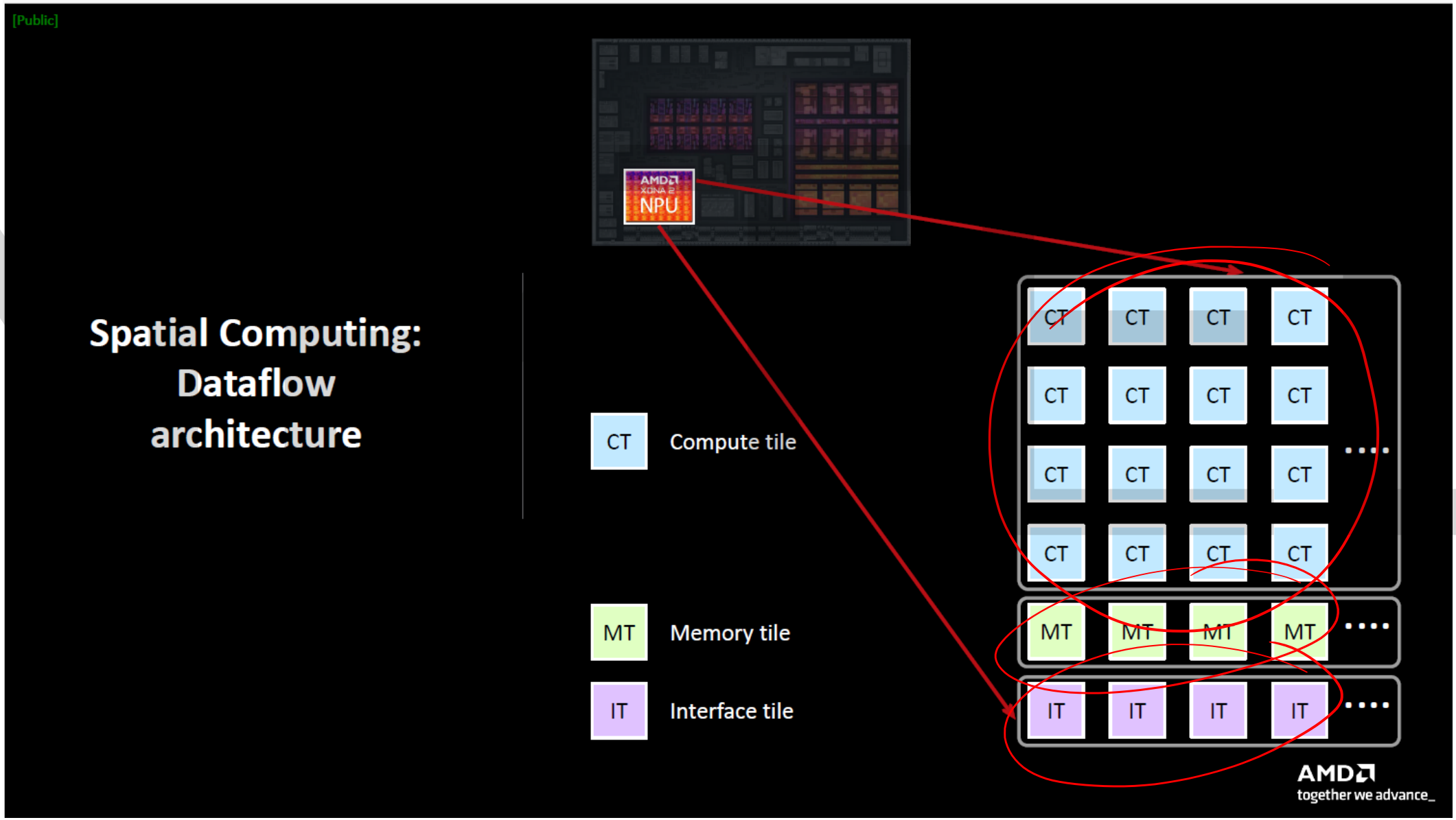
Operation Type Profiling

	Add	Sub	Mul	Div	Pow	Softmax	MatMul	CrossEntropy	MaxPoolGrad	Sum	Conv2D	Conv2DBackFilter	Conv2DBackInput	RandomNormal	ApplyAdam	ApplyRMSProp	Transpose	Tile	Select	Pad	Reshape	Shape
	A						B	C			D			E	F		G					
seq2seq	3	2	35	0	0	0	32	0	0	2	0	0	0	0	0	0	3	20	0	1	0	0
memnet	2	1	33	1	0	4	12	2	0	9	0	0	0	0	5	0	9	13	0	0	1	1
speech	0	0	0	0	0	0	89	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0
autoenc	3	0	6	0	5	0	58	0	0	2	0	0	0	5	8	0	0	9	0	0	0	0
residual	0	0	0	0	0	0	0	0	0	0	33	34	32	0	0	0	0	0	0	0	0	0
vgg	0	0	0	0	0	0	0	0	0	0	35	31	30	0	2	0	0	0	0	0	0	0
alexnet	0	0	0	0	0	0	7	0	3	0	31	26	31	0	0	0	0	0	0	0	0	0
deepq	0	0	0	0	0	0	11	0	0	0	33	27	20	0	0	7	0	0	0	0	0	0

Group	Op Class
A	Elementwise Arithmetic
B	Matrix Operations
C	Reduction and Expansion
D	Convolution
E	Random Sampling
F	Optimization
G	Data Movement

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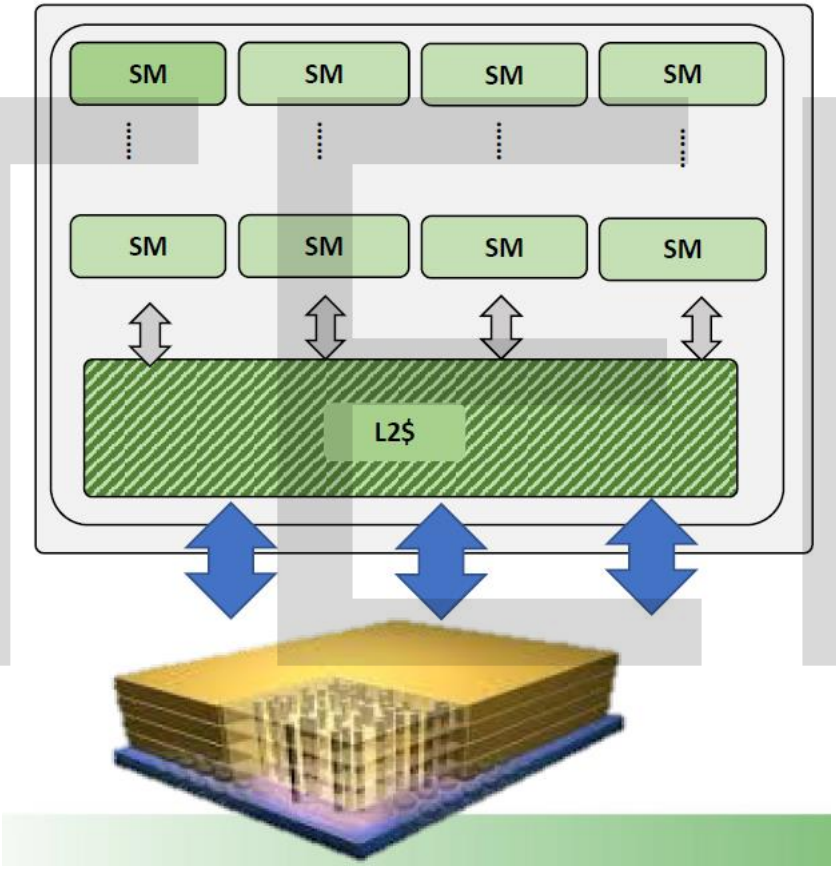
Neural Processing Units (NPU)



Execution Hierarchies in GPUs



SM -> Symmetric Multiprocessor



High Bandwidth Memory (HBM)

Key Takeaways

- Machine Learning (ML) / Artificial Intelligence (AI) algorithms can do wonder. But
  - They are extremely compute and memory intensive
  - Need rigorous optimization of computer system and software to make ML execution possible on the computer (with CPU/GPU)
- Study of system optimizations for ML/AI algorithms is extremely relevant
  - High performance computer shapes the future of AI
  - If there is no system optimization, AI algorithm development will take a major hit
- How to start systems research for ML/AI?
  - Don't be satisfied with high accuracy, think about compute/memory penalty
  - Don't hesitate to make your hands dirty (to dig into architecture, compiler, OS,….)

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Thank You

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  - Kaushik, Utkarsh
- Entire NPTEL team at IISc

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